# Functional Analysis

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## §1 Preliminaries on Topological Vector Spaces

**Lemma 1.1 (Riesz Lemma).** Let $X$ be a normed linear space and $Y \subsetneq X$ a proper closed subspace. Then, for every $0 < \alpha < 1$, there is an $x \in X \setminus Y$ such that $\|x\| = 1$ and $\operatorname{dist}(x, Y) > \alpha$.

## §2 Completeness Arguments

**Theorem 2.1 (Rudin, Exercise 4.26).** Let $X$ and $Y$ be Banach spaces. The set of all surjective bounded linear operators in $\mathscr{B}(X,Y)$ forms an open subset.

*Proof.* Let $T : X \to Y$ be a surjective linear operator. By the open mapping theorem, there is an $r > 0$ such that $B_Y(0,2r) \subseteq T(B_X(0,1))$. If $0 \neq y \in Y$, then $\frac{ry}{\|y\|} \in B_Y(0,2r)$, consequently, there is an $x' \in X$ with $\|x'\| < 1$ and $Tx' = \frac{ry}{\|y\|}$, thus, $x = \frac{\|y\|}{r}x'$ maps to $y$ under $T$. Note that $\|x\| < \frac{\|y\|}{r}$. For the sake of brevity, let $t = 1/r$.

Let $\delta = \frac{1}{2t} > 0$ and $S \in \mathscr{B}(X,Y)$ such that $\|T - S\| < \delta$. We shall show that $S$ is surjective, for which, it would suffice to show that the image of $S$ contains the unit ball of $Y$. Indeed, let $y_0 \in Y$ with $\|y_0\| \leqslant 1$. Choose an $x_0 \in X$ such that $\|x_0\| < t$ and $Tx_0 = y_0$. Setting $y_1 = y_0 - Sx_0$, we have

$$\|y_1\| = \|(T - S)x_0\| \leqslant \delta t.$$

Again, choose $x_1 \in X$ such that $Tx_1 = y_1$ and $\|x_1\| < t\|y_1\| = \delta t^2$. Setting $y_2 = y_1 - Sx_1$, we have

$$\|y_2\| = \|(T - S)x_1\| \leqslant \delta^2 t^2$$

and so on. We have thus constructed two sequences $(x_n)_{n\geqslant 0}$ and $(y_n)_{n\geqslant 0}$ such that

- $Tx_n = y_n$,
- $y_{n+1} = y_n - Sx_n$ for $n \geqslant 0$, and
- $\|x_n\| < \delta^n t^{n+1}$ and $\|y_n\| \leqslant \delta^n t^n$.

Let $x = \sum_{n=0}^{\infty} x_n$, which converges since $\sum_{n=0}^{\infty} \|x_n\|$ does. Hence,

$$Sx = \lim_{n\to\infty} \sum_{i=0}^{n} Sx_i = \sum_{i=0}^{\infty} y_i - y_{i+1} = y_0,$$

thereby completing the proof. ∎

## §3 The Hahn-Banach Theorems

**Lemma 3.1 (Dominated Extension Theorem).** Let $X$ be a real vector space with a subspace $M$. Suppose $p : X \to \mathbb{R}$ satisfies

$$p(x + y) \leqslant p(x) + p(y) \quad \text{and} \quad p(tx) = tp(x) \quad \forall x, y \in M,\ \forall t \geqslant 0.$$

Let $f : X \to \mathbb{R}$ be a linear functional such that $f(x) \leqslant p(x)$ for all $x \in M$. Then, there is a linear functional $\Lambda : X \to \mathbb{R}$ such that $\Lambda x = f(x)$ for all $x \in M$ and

$$-p(-x) \leqslant \Lambda x \leqslant p(x) \quad \forall x \in X.$$

*Proof.* If $M = X$, then there is nothing to prove. Suppose now that $M$ is a proper subspace of $X$ and choose $x_1 \in X \setminus M$. For $x, y \in M$, we have

$$f(x) + f(y) = f(x+y) \leqslant p(x+y) \leqslant p(x - x_1) + p(y + x_1),$$

and hence,

$$f(x) - p(x - x_1) \leqslant -f(y) + p(y + x_1) \quad \forall x, y \in M.$$

Let $\alpha$ denote the supremum of the left hand side in the above inequality as $x$ ranges over $M$. Note that $\alpha$ is finite as the left hand side is always bounded above by $p(x_1)$. Let $M_1 = M + \mathbb{R}x_1$ and define $f_1 : M_1 \to \mathbb{R}$ by

$$f_1(m + \lambda x_1) = f(m) + \lambda\alpha;$$

in particular, $f_1(x_1) = \alpha$. Note that for $\lambda \neq 0$,

$$\begin{aligned} f_1(m + \lambda x_1) &= |\lambda| f_1(|\lambda|^{-1} m + \operatorname{sgn}(\lambda) x_1) \\ &= |\lambda| f(|\lambda|^{-1} m) + \lambda\alpha \\ &\leqslant |\lambda| \left( p(|\lambda|^{-1} m + \operatorname{sgn}(\lambda) x_1) - \operatorname{sgn}(\lambda)\alpha \right) \\ &= p(m + \lambda x_1). \end{aligned}$$

This furnishes an extension $f_1 : M_1 \to \mathbb{R}$ such that $f_1(y) \leqslant p(y)$ for all $y \in M_1$. One can then extend this, using Zorn's Lemma, to $\Lambda : X \to \mathbb{R}$ such that $\Lambda x \leqslant p(x)$ for all $x \in X$. We then have

$$-p(-x) \leqslant -\Lambda(-x) = \Lambda x \leqslant p(x),$$

thereby commpleting the proof. ■

**THEOREM 3.2 (HAHN-BANACH EXTENSION THEOREM).** Let $M$ be a subspace of a vector space (real or complex) $X$, $p$ a semi-norm on $X$, and $f$ a linear functional on $M$ such that $|f(x)| \leqslant p(x)$ for all $x \in M$. Then $f$ extends to a linear functional $\Lambda$ on $X$ satisfying $|\Lambda x| \leqslant p(x)$ for all $x \in X$.

*Proof.* Suppose first that the field of scalars is $\mathbb{R}$. Due to the preceding lemma, $f$ can be extended to $\Lambda : X \to \mathbb{R}$ satisfying

$$-p(x) = -p(-x) \leqslant \Lambda x \leqslant p(x) \quad \forall x \in X,$$

that is, $|\Lambda x| \leqslant p(x)$.

Next, suppose the field of scalars is $\mathbb{C}$. Let $u = \Re f$. Due to the first part of the proof, $u$ can be extended to a linear functional $U : X \to \mathbb{R}$ satisfying $|Ux| \leqslant p(x)$ for all $x \in X$. Define $\Lambda : X \to \mathbb{C}$ by

$$\Lambda x = u(x) - iu(ix) \quad \forall x \in X.$$

We contend that $\Lambda$ is the desired functional. Let $x \in X$ and choose an $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ such that $\alpha\Lambda x = |\Lambda x|$. Hence,

$$|\Lambda x| = \alpha\Lambda x = \underbrace{\Lambda(\alpha x) = U(\alpha x)}_{\text{because LHS } \in \mathbb{R}_{\geqslant 0}} \leqslant p(\alpha x) = p(x).$$

This completes the proof. ■

**COROLLARY.** Let $X$ be a normed linear space and $M$ a subspace of $X$. Suppose $f : M \to \mathbb{K}$ is a bounded linear functional, then there exists a bounded linear functional $\Lambda : X \to \mathbb{K}$ extending $f$. Further, $\|f\| = \|\Lambda\|$

*Proof.* Invoke the preceding result with $p(x) = \|f\|\|x\|$. I ■

**THEOREM 3.3 (HAHN-BANACH SEPARATION THEOREM).** Suppose $A$ and $B$ are disjoint convex subsets of a topological vector space $X$.

(a) If $A$ is open, there exist $\Lambda \in X^*$ and $\gamma \in \mathbb{R}$ such that

$$\Re\Lambda x < \gamma \leqslant \Re\Lambda y \quad \forall x \in A,\ y \in B.$$

(b) If $A$ is compact, $B$ is closed, and $X$ is locally convex, there exist $\Lambda \in X^*$ and $\gamma_1, \gamma_2 \in \mathbb{R}$ such that

$$\Re\Lambda x < \gamma_1 < \gamma_2 < \Re\Lambda y \quad \forall x \in A,\ y \in B.$$

*Proof.* We first prove this theorem when the scalar field is assumed to be $\mathbb{R}$.

(a) Fix points $a_0 \in A$, $b_0 \in B$. Set $x_0 = b_0 - a_0$ and $C = A - B + x_0$. Then, $C$ is a convex neighborhood of 0 in $X$, and thus, admits a Minkowski functional, $p : X \to \mathbb{R}$ which is subadditive and $p(tx) = tp(x)$ for all $t \geqslant 0$. Further, since $A \cap B = \emptyset$, $x_0 \notin C$, whence $p(x_0) \geqslant 1$.

Define a linear functional $f : \mathbb{R}x_0 \to \mathbb{R}$ by $f(\lambda x_0) = \lambda$ and using the Dominated Extension Theorem, extend this to a functional $\Lambda : X \to \mathbb{R}$ such that

$$-p(-x) \leqslant \Lambda x \leqslant p(x) \quad \forall x \in X.$$

Let $D = C \cap (-C)$, which is a symmetric convex neighborhood of the origin. For any $x \in D$, it is easy to see that $p(x) \leqslant 1$, whence

$$-1 \leqslant -p(-x) \leqslant \Lambda x \leqslant p(x) \leqslant 1,$$

and hence, $\Lambda$ is a continuous linear functional.

Now, for $a \in A$ and $b \in B$,

$$\Lambda a - \Lambda b = \Lambda(a - b) = \Lambda(a - b + x_0) - 1 \leqslant p(a - b + x_0) - 1 < 0,$$

since $a - b + x_0 \in C$. Hence, $\Lambda a < \Lambda b$ for every $a \in A$ and $b \in B$. Finally, since $\Lambda(A)$ and $\Lambda(B)$ are disjoint convex subsets of $\mathbb{R}$, both must be intervals with the former to the left of the latter. Further, since the former is an open subset of $\mathbb{R}$, we immediately obtain the desired conclusion.

(b) There is a convex, balanced neighborhood $V$ of the origin in $X$ such that $(A + V) \cap (B + V) = \emptyset$. Set $C = A + V$, which is a convex open subset of $X$, disjoint from $B$. Due to part (a), there is a linear functional $\Lambda$ such that $\Lambda(C)$ is to the left of $\Lambda(B)$ and $\Lambda(A)$ sits as a compact interval inside $\Lambda(C)$. The conclusion now is immediate.

We now suppose that the field of scalars is $\mathbb{C}$; whence $X$ is also a topological $\mathbb{R}$-vector space. In both parts (a) and (b), we were able to obtain an $\mathbb{R}$-linear functional, continuous on $X$ when viewed as a $\mathbb{R}$-TVS and separating the two sets as desired. Define the $\mathbb{C}$-linear functional $\Lambda x = u(x) - iu(ix)$ and note that this has the desired separation properties too. ■

**COROLLARY.** If $X$ is an LCTVS, then $X^*$ separates points on $X$.

*Proof.* Let $p, q \in X$. Use Theorem 3.3 (b) with $A = \{p\}$ and $B = \{q\}$. ■

**THEOREM 3.4.** Let $M$ be a proper closed subspace of a locally convex topological vector space, and $x_0 \in X \setminus M$. There exists a linear functional $\Lambda \in X^*$ such that $\Lambda x_0 = 1$ and $\Lambda x = 0$ for all $x \in M$.

*Proof.* Using Theorem 3.3(b) with $A = \{x_0\}$ and $B = M$, there is a $\Lambda \in X^*$ and $\gamma_1, \gamma_2 \in \mathbb{R}$ such that

$$\Re \Lambda x_0 < \gamma_1 < \gamma_2 < \Re \Lambda y \quad \forall y \in M.$$

Since $\Lambda(0) = 0$ and $0 \in M$, we must have that $\Lambda x_0 \neq 0$. Further, since $\lambda y \in M$ for every $\lambda \in \mathbb{K}$, the only way $\Re(\lambda \Lambda y) > \gamma_2$ for every $\lambda \in \mathbb{K}$ is if $\Lambda$ vanishes on $M$. Dividing $\Lambda$ by $\Lambda x_0$, we have our desired conclusion. ■

**Corollary.** Let $X$ be an LCTVS and $M \subseteq X$ a subspace. Suppose $f : M \to \mathbb{K}$ is a continuous linear functional, then there is a $\Lambda \in X^*$ such that $\Lambda|_M = f$.

*Proof.* ■

## §4 Weak and Weak* Topologies

**Lemma 4.1.** Let $X$ be a $\mathbb{K}$-vector space and $\Lambda_1, \ldots, \Lambda_n, \Lambda$ be linear functionals on $X$ and set

$$N = \{x \in X : \Lambda_i x = 0, \ \forall 1 \leqslant i \leqslant n\}.$$

The following are equivalent:

(a) There are scalars $\alpha_1, \ldots, \alpha_n \in \mathbb{K}$ such that

$$\Lambda = \alpha_1 \Lambda_1 + \cdots + \alpha_n \Lambda_n.$$

(b) There exists $0 < \gamma < \infty$ such that

$$|\Lambda x| \leqslant \gamma \max_{1 \leqslant i \leqslant n} |\Lambda_i x| \quad \forall x \in X.$$

(c) $\Lambda x = 0$ for every $x \in N$.

*Proof.* $(a) \implies (b) \implies (c)$ is trivial. It remains to show that $(c) \implies (a)$. Consider the map $\Phi : X \to \mathbb{K}^n$ given by

$$\Phi(x) = (\Lambda_1 x, \ldots, \Lambda_n x)$$

and let $Y \subseteq \mathbb{K}^n$ be its image. Define $\Psi : Y \to \mathbb{K}$ by

$$\Psi(\Phi(x)) = \Lambda x.$$

That this is well-defined follows from the fact that $N \subseteq \ker \Lambda$. Since we are in a finite-dimensional space, the map $\Psi$ can be extended to a linear map $\Psi : \mathbb{K}^n \to \mathbb{K}$, which must be of the form

$$(y_1, \ldots, y_n) \mapsto \alpha_1 y_1 + \cdots + \alpha_n y_n.$$

It then follows that $\Lambda = \alpha_1 \Lambda_1 + \cdots + \alpha_n \Lambda_n$. ■

**Definition 4.2.** Let $X$ be a set and

$$\mathscr{F} = \{f : X \to Y_f\}$$

a collection of functions. The *$\mathscr{F}$-topology* on $X$ is defined to be the coarsest topology such that every $f \in \mathscr{F}$ is continuous.

The set $\mathscr{F}$ is said to *separate points* if for each pair $p \neq q$ in $X$, there is an $f \in \mathscr{F}$ such that $f(p) \neq f(q)$.

**REMARK 4.3.** The $\mathscr{F}$-topology is more explicitly the topology generated by

$$\{f^{-1}(U) \colon U \subseteq Y_f \text{ is open}, \ f \in \mathscr{F}\}.$$

**PROPOSITION 4.4.** If $\mathscr{F}$ is a separating family of functions on a space $X$, and each $Y_f$ is Hausdorff, then the $\mathscr{F}$-topology on $X$ is Hausdorff.

*Proof.* Let $p \neq q$ be points in $X$ and choose $f \in \mathscr{F}$ such that $f(p) \neq f(q)$. Then, there are disjoint neighborhoods $U$ and $V$ of $f(p)$ and $f(q)$ respectively in $Y_f$. Since each $f$ is continuous, $f^{-1}(U)$ and $f^{-1}(V)$ are disjoint neighborhoods of $p$ and $q$ in the $\mathscr{F}$-topology. ■

**PROPOSITION 4.5.** If $X$ is a compact topological space and $\mathscr{F}$ is a countable family of continuous separating real-valued functions on $X$, then $X$ is metrizable.

*Proof.* Let $\mathscr{F} = \{f_n : n \geqslant 1\}$. We may suppose without loss of generality that $\|f\|_\infty \leqslant 1$ for each $f \in \mathscr{F}$. It is not hard to check that the function $d : X \times X \to \mathbb{R}$ given by

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n} |f_n(x) - f_n(y)|$$

is a metric inducing the topology on $X$. ■

**THEOREM 4.6.** Let $X$ be a $\mathbb{K}$-vector space and $X'$ a vector space of linear functionals on $X$ that separates points. The $X'$-topology $\tau'$ on $X$ makes it a locally convex topological vector space whose dual is $X'$.

*Proof.* Due to Proposition 4.4, $\tau'$ is Hausdorff. Note that the topology is generated by the set

$$\{\Lambda^{-1}(U) \colon \Lambda \in X', \ U \subseteq \mathbb{K} \text{ is open}\}.$$

Hence, a base for the topology is given by finite intersections of elements of the above form. Thus, is generated by intersections of the form

$$\Lambda_1^{-1}(U_1) \cap \cdots \cap \Lambda_n^{-1}(U_n),$$

where $U_1, \ldots, U_n \subseteq \mathbb{K}$ are open sets. It immediately follows that this base is translation invariant whence, the entire topology is translation invariant. A local base at 0 is given by open sets of the above form, such that $0 \in U_i$ for $1 \leqslant i \leqslant n$. We can further refine this local base by choosing open sets of the form

$$V(\Lambda_1, \ldots, \Lambda_n; \varepsilon_1, \ldots, \varepsilon_n) = \{x \in X : |\Lambda_i x| \leqslant \varepsilon_i, \ 1 \leqslant i \leqslant n\}.$$

Further, from this description, it is not hard to see that $\alpha V$ is a basic open set whenever $\alpha > 0$ and $V$ a basic open set.

Now that we have established a local base for $\tau'$, we show that $(X, \tau')$ is indeed a topological vector space. That $\tau'$ is locally convex immediately follows from the above description of a local base. Next, we show that addition is continuous, for which it suffices to show continuity at $(0,0) \in X \times X$. Let $U$ be a neighborhood of 0 in $X$, then $U$ contains a basic open set $V$ of the above form. Since $\frac{1}{2}V + \frac{1}{2}V \subseteq V$, we see that addition is continuous.

To see that scalar multiplication is continuous, let $x \in X$, $\alpha \in \mathbb{K}$ and $x + V$ a neighborhood of $x$. We may suppose that $V$ is a basic open set of the above form. Since $V$ is absorbing, there is an $s > 0$ such that $x \in sV$. Choose $r$ sufficiently small so that $r(r+s) + r|\alpha| < 1$. Then, if $y \in x + rV$, and $|\beta - \alpha| < r$,

$$\beta y - \alpha x = (\beta - \alpha)y + \alpha(y - x) \in r(r+s)V + |\alpha| rV \subseteq V,$$

since $y \in (r+s)V$. Hence, scalar multiplication is continuous and $(X, \tau')$ is a locally convex topological vector space.

Finally, let $\Lambda$ be a continuous linear functional on $X$ and consider a basic open set $V(\Lambda_1, \ldots, \Lambda_n, \varepsilon_1, \ldots, \varepsilon_n)$ such that $|\Lambda x| < 1$ on $V$. Thus, there is a $\gamma > 0$ such that

$$|\Lambda x| \leqslant \gamma \max_{1 \leqslant i \leqslant n} |\Lambda_i x|$$

whence, $\Lambda$ is a linear combination of the $\Lambda_i$. ■

**Definition 4.7.** Let $X$ be a topological vector space whose dual $X^*$ separates points on $X$ (this is true in particular for locally convex TVSs). Then the $X^*$-topology on $X$ is called the *weak topology* and is denoted by $(X, \tau_w)$ or $X_w$.

Obviously the weak topology is coarser than the original topology. A set $E \subseteq X$ is said to be *weakly bounded* if it is bounded in the weak topology. Similarly, a sequence $(x_n)$ is said to be *weakly convergent* to $x$ if it converges in the weak topology. Since the weak topology is Hausdorff, the limit of any weakly convergent sequence is unique.

**Proposition 4.8.** Let $X$ be a topological vector space on which $X^*$ separates points. Then

(a) $X_w$ is a locally convex topological vector space.

(b) A set $E \subseteq X$ is weakly bounded if and only if every $\Lambda \in X^*$ is bounded on $E$.

(c) A sequence $(x_n)$ is weakly convergent to $x$ if and only if $\Lambda x_n \to \Lambda x$ for every $\Lambda \in X^*$.

*Proof.* All three assertions are trivial. ■

**Proposition 4.9.** Let $X$ be a locally convex topological vector space and $E \subseteq X$ a convex subset. Then the weak closure $\overline{E}_w$ is the same as the original closure $\overline{E}$.

*Proof.* Since the weak topology is coarser than the original topology, $\overline{E} \subseteq \overline{E}_w$. Now, let $x_0 \in X \setminus \overline{E}$. Due to the Hahn-Banach Separation Theorem, there is an $\Lambda \in X^*$ and $\gamma_1, \gamma_2 \in \mathbb{R}$ such that

$$\Re \Lambda x_0 < \gamma_1 < \gamma_2 < \Re \Lambda y \quad \forall y \in \overline{E} \supseteq E.$$

Thus, there is a weak neighborhood of $x_0$ not intersecting $E$, consequently, $x_0 \notin \overline{E}_w$. This completes the proof. ■

**Theorem 4.10.** Suppose $X$ is an infinite-dimensional normed linear space. Then the weak topology on $X$ is not metrizable.

*Proof.* We shall show that the weak topology $(X, w)$ is not first-countable whence the conclusion would follow. Suppose not, then there is a local base $\{U_n\}$ at 0. For each $n \geqslant 1$, there is a finite subset $F_n \subseteq X^*$ and $\varepsilon_n > 0$ such that

$$V_n = \{x \in X : |f(x)| < \varepsilon_n, \ \forall f \in F_n\}.$$

We contend that

$$X^* = \bigcup_{n \geqslant 1} \operatorname{span}(F_n).$$

Indeed, let $g \in X^*$ and

$$U = \{x \in X : |g(x)| < 1\}.$$

There is an index $n \geqslant 1$ such that $V_n \subseteq U$. Now, if $x$ is in $\bigcap_{f \in F_n} \ker f$, then so is $\lambda x$ for every $\lambda \in \mathbb{K}$, consequently, $\lambda x \in V_n$ and hence, $|\lambda||g(x)| < 1$ for every $\lambda \in \mathbb{K}$. This forces $g(x) = 0$, that is,

$$\bigcap_{f \in F_n} \ker f \subseteq \ker g,$$

which, in light of Lemma 4.1 gives $g \in \operatorname{span}(F_n)$, proving our claim.

It follows that $X^*$ has at most countable dimension and since $X$ is infinite-dimensional, so is $X^*$, but this is absurd, since $X^*$ is a Banach space. ■

**Definition 4.11.** Let $X$ be a topological vector space and $X^*$. The evaluation functionals induced by $X$ form a separating vector space of functionals. The $X$-topology induced on $X^*$ by these functionals is called the *weak* topology*.

**Theorem 4.12 (Banach-Alaoglu).** Let $X$ be a topological vector space and $V$ a neighborhood of 0. The *polar* of $V$:

$$K = \{\Lambda \in X^* : |\Lambda x| \leqslant 1,\ \forall x \in V\} \subseteq X^*$$

is weak*-compact.

*Proof.* Since $V$ is a neighborhood of the origin, it is absorbing and hence, for each $x \in X$, there is $\gamma(x) > 0$ such that $x \in \gamma(x)V$. For $x \in V$, choose $\gamma(x) \leqslant 1$. Let $D_x$ denote the compact set

$$D_x = \left\{z \in \mathbb{K} : |z| \leqslant \gamma(x)\right\}, \tag{1}$$

and

$$P = \prod_{x \in X} D_x,$$

which is compact due to Tychonoff's Theorem. Further, for each $\Lambda \in K$ and $x \in X$, since $x/\gamma(x) \in V$, we have $|\Lambda x| \leqslant |\gamma(x)|$, consequently, the element $(\Lambda x)_{x \in X}$ is an element of $P$. Thus, we can identify $K$ with a subset of $P$. Henceforth, we shall denote elements of $P$ as functions $f : X \to \mathbb{K}$. We shall show that:

(i) the subspace topology $K$ inherits from $P$ and the weak*-topology on $K$ are the same,

(ii) with respect to the subspace topology, $K$ is closed in $P$;

whence it follows that $K$ is compact.

Let $\Lambda_0 \in K$ and consider a basic open set in the weak*-topology centered at $\Lambda_0$ of the form

$$W = \left\{\Lambda \in X^* : |\Lambda x_i - \Lambda_0 x_i| < \varepsilon,\ 1 \leqslant i \leqslant n\right\}.$$

In the product topology on $P$, the following set is open

$$V = \{f \in P : |f(x_i) - \Lambda_0 x_i| < \varepsilon,\ 1 \leqslant i \leqslant n\}.$$

It is not hard to see that $W \cap K = V \cap K$. This shows that the subspace topology induced on $K$ by the product topology is finer than that induced by the weak*-topology.

On the other hand, choose any open set in the product topology in $P$ intersecting $K$ and choose an element $\Lambda_0$ in the intersection. The aforementioned open set contains one of the form $V$ as above and since $W \cap K = V \cap K$, we see that the weak*-topology is finer than the subspace topology. This shows that the two topologies are the same.

Finally, we must show that $K$ is closed in $P$. Let $f_0 \in \overline{K}$, $x, y \in X$ and $\alpha, \beta \in \mathbb{K}$. We contend that $f_0(\alpha x + \beta y) = \alpha f_0(x) + \beta f_0(y)$. Let $\varepsilon > 0$ and

$$V = \left\{ f \in P : |f(z) - f_0(z)| < \varepsilon,\ z \in \{x, y, \alpha x + \beta y\} \right\}.$$

There is some $f \in K \cap V$. Then,

$$\begin{aligned}
&|f_0(\alpha x + \beta y) - \alpha f(x) - \beta f(y)| \leqslant \\
&|f_0(\alpha x + \beta y) - f(\alpha x + \beta y)| + |\alpha f(x) - \alpha f_0(x)| + |\beta f(y) - \beta f_0(y)| \\
&\leqslant (|\alpha| + |\beta| + 1)\varepsilon.
\end{aligned}$$

Since the above inequality holds for all $\varepsilon > 0$, we have that $f_0$ is linear. Further, by construction, $f_0$ is bounded by 1 on $V$, since $\gamma(x) \leqslant 1$ for all $x \in V$ and hence, $f_0 \in X^*$. It follows that $f_0 \in K$ and hence, $K$ is closed in $P$, thereby completing the proof. ■

**Proposition 4.13 (Rudin, Exercise 3.11).** Let $X$ be an infinite dimensional Fréchet space. Then $X^*$ with the weak*-topology is of the first category in itself.

*Proof.* Let $V_n = B(0, 1/n) \subseteq X$ and let $K_n$ denote their respective polars, that is

$$K_n = \{\Lambda \in X^* : |\Lambda x| \leqslant 1,\ \forall x \in V_n\}.$$

First, we claim that $X^* = \bigcup_{n=1}^{\infty} K_n$. Indeed, for any $\Lambda \in X^*$, note that the open set $\Lambda^{-1}(B_{\mathbb{K}}(0,1))$ contains some $V_n$ and hence, $\Lambda \in K_n$.

It remains to now show that these have empty interior. Indeed, suppose $K_N$ has nonempty interior for some $N \in \mathbb{N}$. Since $K_N$ is convex, symmetric, so is its interior. Thus, we have that 0 lies in the interior of $K_N$. As a result, there is an $\varepsilon > 0$ and $x_1, \ldots, x_n \in X$ such that

$$W = \left\{ \Lambda \in X^* : |\Lambda x_i| < \varepsilon,\ 1 \leqslant i \leqslant n \right\} \subseteq K_N.$$

Since $K_N$ is compact, it is bounded and hence, so is $W$. But since $X^*$ is infinite-dimensional too, so is $\bigcap_{i=1}^{n} \ker \mathrm{ev}_{x_i} \subseteq W$ which is contained in a bounded set, whence, must be the trivial subspace.

Next, for any $x \in X$, note that

$$\bigcap_{i=1}^{n} \ker \mathrm{ev}_{x_i} = \{0\} \subseteq \ker \mathrm{ev}_x,$$

thus $x$ is a linear combination of the $x_i$'s, that is, $X$ is finite-dimensional, a contradiction. This completes the proof. ■

## §§ The Krein-Milman Theorem

**Definition 4.14.** A subset $E$ of a topological vector space $X$ is said to be *totally bounded* if to every neighborhood $V$ of 0 in $X$ corresponds a finite set $F$ such that $E \subseteq F + V$.

**Remark 4.15.** Note that we can require that $F \subseteq E$. Indeed, let $V$ be a neighborhood of 0 and choose a neighborhood $W$ of 0 such that $W + W \subseteq V$. There is a finite set $F \subseteq X$ such that $E \subseteq F + W$. For each $f \in F$ such that $(f + W) \cap E \neq \emptyset$, choose some $e$ in the intersection. For any $w \in W$, we have $f + w - e = (f - e) + w \in W + W \subseteq V$. Hence, $f + W \subseteq e + V$. The collection of all such $e$'s, say $\widetilde{F}$ is such that $E \subseteq \widetilde{F} + W$

**Theorem 4.16.** (a) If $A_1, \ldots, A_n$ are compact convex sets in a topological vector space $X$, then $\mathrm{co}(A_1 \cup \cdots \cup A_n)$ is compact.

(b) If $X$ is an LCTVS and $E \subseteq X$ is totally bounded, then $\mathrm{co}(E)$ is totally bounded.

(c) If $X$ is a Fréchet space and $K \subseteq X$ is compact, then $\overline{\mathrm{co}}(X)$ is compact.

*Proof.* (a) Let

$$\Delta = \left\{(s_1, \ldots, s_n) \in \mathbb{R}^n : s_1 + \cdots + s_n = 1,\ s_i \geqslant 0\ \forall 1 \leqslant i \leqslant n\right\}.$$

Let $A = A_1 \times \cdots \times A_n$ and define the map $f : \Delta \times A \to X$ by

$$f(s, a) = s_1 a_1 + \cdots + s_n a_n.$$

This is a continuous map since addition and scalar multiplication are continuous on $X$. Put $K = f(S \times A)$. Then, $K$ is compact and is contained in $\mathrm{co}(A_1 \cup \cdots \cup A_n)$.

We shall show that $K = \mathrm{co}(A_1 \cup \cdots \cup A_n)$, for which is suffices to show that $K$ is convex (since each $A_i$ is contained in $K$). Indeed, let $\alpha, \beta > 0$ with $\alpha + \beta = 1$. Then, for $(s, a), (t, b) \in S \times A$, we have

$$\alpha \sum_{i=1}^{n} s_i a_i + \beta \sum_{i=1}^{n} t_i b_i = \sum_{i=1}^{n} (\alpha s_i + \beta t_i) \cdot \frac{\alpha s_i a_i + \beta t_i b_i}{\alpha s_i + \beta t_i} = f(u, c),$$

where $u = \alpha s + \beta t$ and

$$c_i = \frac{\alpha s_i a_i + \beta t_i b_i}{\alpha s_i + \beta t_i} \in A_i,$$

and we are done.

(b) Let $U$ be a neighborhood of 0 in $X$ and choose a convex, balanced neighborhood $V$ of 0 in $X$ such that $V + V \subseteq U$. There is a finite set $F \subseteq X$ such that $E \subseteq F + V$, whence $E \subseteq \mathrm{co}(F) + V$. Since the latter is convex, we have $\mathrm{co}(E) \subseteq \mathrm{co}(F) + V$.

Due to part (a), $\mathrm{co}(F)$ is compact. The collection $\{f + V : f \in \mathrm{co}(F)\}$ is an open cover of $\mathrm{co}(F)$ and hence, admits a finite subcover, $\mathrm{co}(F) \subseteq F_1 + V$ for some $F_1 \subseteq X$. Therefore,

$$\mathrm{co}(E) \subseteq F_1 + V + V \subseteq F_1 + U,$$

that is, $\mathrm{co}(E)$ is totally bounded.

(c) Due to part (b), $\mathrm{co}(K)$ is totally bounded. Thus, its closure is totally bounded and complete, whence compact. ■

**LEMMA 4.17 (CARATHÉODORY).** If $E \subseteq \mathbb{R}^n$ and $x \in \mathrm{co}(E)$, then $x$ lies in the convex hull of of some subset of $E$ which contains at most $n + 1$ points.

*Proof.* We shall show that if $k > n$ and $x = \sum_{i=1}^{k+1} t_i x_i$ is a convex combination for some $x_i \in \mathbb{R}^n$, then $x$ is a convex combination of some $k$ of these vectors. This is enough to prove the statement of the theorem.

We may suppose without loss of generality that $t_i > 0$ for $1 \leqslant i \leqslant k + 1$. Consider the linear map $\mathbb{R}^{k+1} \to \mathbb{R}^{n+1}$ given by

$$(a_1, \ldots, a_{k+1}) \mapsto \left(\sum_{i=1}^{k+1} a_i x_i, \sum_{i=1}^{k+1} a_i\right).$$

The kernel of this map must be nontrivial and hence, there exists $(a_1, \ldots, a_{k+1}) \in \mathbb{R}^{k+1}$ with some $a_i \neq 0$, so that $\sum_{i=1}^{k+1} a_i x_i = 0$ and $\sum_{i=1}^{k+1} a_i = 0$. Set

$$|\lambda| = \min_{1 \leqslant i \leqslant k+1} \frac{t_i}{|a_i|},$$

which is finite, since $a_i \neq 0$ for some $1 \leqslant i \leqslant k+1$. Choose the sign of $\lambda$ so that $\lambda a_j = \lambda_j$ for some $1 \leqslant j \leqslant k+1$. Set $c_i = t_i - \lambda a_i \geqslant 0$. Then,

$$\sum_{i=1}^{k+1} c_i x_i = \sum_{i=1}^{k+1} t_i x_i - \lambda \sum_{i=1}^{k+1} a_i x_i = x,$$

and

$$\sum_{i=1}^{k+1} c_i = \sum_{i=1}^{k+1} t_i - \lambda \sum_{i=1}^{k+1} a_i = 1.$$

Note that $c_j = 0$ and hence, we have written $x$ as a convex combination of some $k$ of the $x_i$'s. ■

**PROPOSITION 4.18.** If $K \subseteq \mathbb{R}^n$ is compact, then so is $\mathrm{co}(K)$.

*Proof.* Let

$$\Delta = \{(s_1, \ldots, s_{n+1}) \in \mathbb{R}^{n+1} : s_1 + \cdots + s_{n+1} = 1,\ s_i \geqslant 0\ \forall 1 \leqslant i \leqslant n+1\}.$$

Due to Carathéodory's lemma, it follows that $x \in \mathrm{co}(K)$ if and only if $x$ is a linear combination of some $n+1$ elements of $K$. Thus, the map $\Delta \times K^{n+1} \to \mathbb{R}^n$ given by

$$(t, x_1, \ldots, x_{n+1}) \mapsto t_1 x_1 + \cdots + t_{n+1} x_{n+1}$$

is continuous and its image is $\mathrm{co}(K)$. This completes the proof. ■

**DEFINITION 4.19.** Let $X$ be a $\mathbb{K}$-vector space and $K \subseteq X$. A non-empty set $S \subseteq K$ is called an *extreme set* of $K$ if whenever $x, y \in K$, $0 < t < 1$ such that $(1-t)x + ty \in S$, then $x, y \in S$.

The *extreme points* of $K$ are the extreme sets that are singletons. The set of all extreme points of $K$ is denoted by $E(K)$.

**LEMMA 4.20.** Let $X$ be a topological vector space on which $X^*$ separates points. Suppose $A, B$ are disjoint, nonempty, compact, convex sets in $X$. Then there exists $\Lambda \in X^*$ such that

$$\sup_{x \in A} \Re \Lambda x < \inf_{y \in B} \Re \Lambda y.$$

*Proof.* Topologize $X$ with the weak topology, which is coarser than the original topology, and hence, $A, B$ are compact. Now, use the Hahn-Banach separation theorem and the fact that $(X_w)^* = X^*$. ■

**THEOREM 4.21 (KREIN-MILMAN).** Let $X$ be a topological vector space on which $X^*$ separates points. If $K \subseteq X$ is a nonempty compact convex set in $X$, then $K = \overline{\mathrm{co}}(E(K))$.

*Proof.* Let $\mathscr{P}$ denote the poset of all nonemtpy compact extreme sets of $K$ ordered by inclusion. Note that $\mathscr{P}$ is nonempty, since $K \in \mathscr{P}$. We make the following two observations about $\mathscr{P}$:

(a) If $S \neq \emptyset$, is an intersection of elements of $\mathscr{P}$, then $S \in \mathscr{P}$.

(b) If $S \in \mathscr{P}$, $\Lambda \in X^*$ and $\mu = \max_{x \in S} \Re \Lambda x$, then

$$S_\Lambda = \{x \in S : \Re \Lambda x = \mu\} \in \mathscr{P}.$$

Observation (a) is obvious. As for (b), first note that $S_\Lambda$ is closed in $S$, and hence, in $K$, thus, is compact. Now, suppose $x, y \in K$ and $t > 0$ such that $tx + (1-t)y \in S_\Lambda \subseteq S$. Since $S$ is an extreme set of $K$, $x, y \in S$, consequently, $\Re\Lambda x, \Re\Lambda y \leqslant \mu$ and

$$\mu = \Re\Lambda(tx + (1-t)y) \leqslant t\mu + (1-t)\mu = \mu,$$

whence $x, y \in S_\Lambda$, thereby proving (b).

Choose some $S \in \mathscr{P}$ and let $\mathscr{P}'$ be the sub-poset of all members of $\mathscr{P}$ that are contained in $S$. Let $\Omega$ be a maximal chain in $\mathscr{P}'$ and let $M$ denote the intersection of all elements of $\Omega$. Since $\Omega$ has the finite intersection property and all sets in $\Omega$ are compact, $M \neq \emptyset$ and is compact.

We contend that $M$ is a singleton. Indeed, since $M_\Lambda \subseteq M$, due to the minimality of $M$, we must have that $M_\Lambda = M$ for every $\Lambda \in X^*$. That is, $\Re\Lambda(x-y) = 0$ for all $x, y \in M$ and $\Lambda \in X^*$. Since $X^*$ separates points on $X$, we must have that $x - y = 0$, that is, $M$ is a singleton.

We have therefore proved that $E(K) \cap S \neq \emptyset$ for every $S \in \mathscr{P}$. Now, since $K$ is convex, $\overline{\mathrm{co}}(E(K)) \subseteq K$, consequently, the former is compact. Suppose now that there is some $x_0 \in K \setminus \overline{\mathrm{co}}(E(K))$. Applying the preceding lemma with $B = \{x_0\}$ and $A = \overline{\mathrm{co}}(E(K))$, there is a $\Lambda \in X^*$ such that

$$\Re\Lambda x_0 > \sup_{y \in \overline{\mathrm{co}}(E(K))} \Re\Lambda y.$$

Then, $K_\Lambda \in \mathscr{P}$ and is disjoint from $\overline{\mathrm{co}}(E(K))$, a contradiction. Thus, $\overline{\mathrm{co}}(E(K)) = K$, thereby completing the proof. ■

## §§ The $BX$-topology

**Definition 4.22.** Let $X$ be a Banach space. The *bounded $X$-topology* or *$BX$-topology* for $X^*$ is the strongest topology which coincides with the weak*-topology on each set $rS_{X^*} = \{x^* \in X^* : \|x^*\| \leqslant r\}$.

That is, a set $U \subseteq X^*$ is open if and only if $U \cap rS_{X^*}$ is (relatively) open in $rS_{X^*}$ for each $r \geqslant 0$.

**Lemma 4.23.** Let $X$ be a Banach space. A *fundamental system of neighborhoods* of the origin for the $BX$-topology on $X^*$ consists of the sets

$$\left\{x^* : |x^*(x)| < 1 \quad \text{for all } x \in A\right\},$$

where $A = (x_n)_{n \geqslant 1}$ is a sequence of elements of $X$ converging to 0.

*Proof.* First, we show that every neighborhood of the above form is open in the $BX$-topology. Indeed, let $r > 0$ and consider the closed ball $rS_{X^*}$ and let $A = (x_n)_{n \geqslant 1}$ be a sequence of elements in $X$ converging to 0. There is a finite subset $A_1 \subseteq A$ consisting of those $x_n$'s with $\|x_n\| \geqslant \frac{1}{r}$. Then it is clear that

$$\left\{x^* \in X^* : |x^*(x)| < 1 \quad \text{for all } x \in A\right\} \cap rS_{X^*} = \left\{x^* \in X^* : |x^*(x)| < 1 \quad \text{for all } x \in A_1\right\} \cap rS_{X^*},$$

is open in the (relative) weak*-topology. Thus our desired set is open in the $BX$-topology.

Next, let $U \subseteq X^*$ be a neighborhood of 0 in the $BX$-topology. We must show that $U$ contains a fundamental neighborhood of the above form. We establish some notation first: for a subset $A \subseteq X$, let $\widetilde{A}$ denote the set

$$\widetilde{A} = \left\{x^* \in X^* : |x^*(x)| \leqslant 1 \quad \text{for all } x \in A\right\}.$$

Note that $\widetilde{A}$ is closed in the weak*-topology (and hence the $BX$-topology) and

$$\widetilde{\bigcup_{i \in I} A_i} = \bigcap_{i \in I} \widetilde{A}_i.$$

Coming back to our proof, since $U$ is open in the $BX$-topology, the set $U \cap S_{X^*}$ is (relatively) open in the weak*-topology, and hence, there exists a finite set $A_1 \subseteq X$ such that

$$\widetilde{A}_1 \cap S_{X^*} \subseteq U \cap S_{X^*} \subseteq U.$$

We now proceed to inductively construct an ascending chain of finite sets $(A_n)_{n\geqslant 1}$ such that $\widetilde{A}_n \cap nS_{X^*} \subseteq U$. Suppose the chain has been constructed up to $A_n$ for some positive integer $n \geqslant 1$. We claim that there exists a finite set $B_n \subseteq X$ such that $B_n \subseteq \overline{B_X\left(0, \frac{1}{n}\right)}$ and

$$\widetilde{A_n \cup B_n} \cap (n+1)S_{X^*} \subseteq U.$$

Suppose not, then the collection of sets

$$\mathscr{F} = \left\{ \widetilde{A_n \cup B} \cap (n+1)S_{X^*} \cap \left(X^* \setminus U\right) : B \subseteq \overline{B_X\left(0, \frac{1}{n}\right)} \text{ a finite set} \right\}$$

has the finite intersection property. Further, note that these sets are closed in the weak*-topology. Due to Theorem 4.12, $(n+1)S_{X^*}$ is compact, and hence, all the sets in $\mathscr{F}$ have a common element, say $x^* \in X^*$. By choice, we have that $|x^*(y)| \leqslant 1$ for every $y \in \overline{B_X\left(0, \frac{1}{n}\right)}$, so that $\|x^*\| \leqslant n$. But this would mean that $x^* \in \widetilde{A}_n \cap nS_{X^*} \cap (X^* \setminus U)$, which is absurd.

Thus, we have shown the existence of a finite set $B_n \subseteq \overline{B_X\left(0, \frac{1}{n}\right)}$ such that $\widetilde{A_n \cup B_n} \cap (n+1)S_{X^*} \subseteq U$. Set $A_{n+1} := A_n \cup B_n$ and continue inductively.

Finally, setting $A := \bigcup_{n=1}^{\infty} A_n$, it is not hard to see that

$$\left\{x^* \in X^* : |x^*(x)| < 1 \quad \text{for all } x \in X\right\} \subseteq U,$$

and that by suitably enumerating $A$, it constitutes a sequence converging to 0, thereby completing the proof. ■

**COROLLARY.** Let $X$ be a Banach space. Then $X^*$ equipped with the $BX$-topology is a locally convex topological vector space.

**THEOREM 4.24.** A linear functional on $X^*$ is continuous in the weak*-topology if and only if it is convergent in the $BX$-topology.

*Proof.* Since the $BX$-topology is finer than the weak*-topology, every linear functional that is continuous in the weak*-topology is automatically continuous in the $BX$-topology. We prove the converse.

Let $\theta : X^* \to \mathbb{K}$ be a continuous linear functional in the $BX$-topology. In view of Lemma 4.23, there exists a fundamental neighborhood of the form

$$\left\{x^* \in X^* : |x^*(x_n)| < 1 \quad \text{for all } n \geqslant 1\right\} \subseteq \theta^{-1}(B_{\mathbb{K}}(0,1)),$$

where $(x_n)_{n\geqslant 1}$ is a sequence converging to 0 (in the norm topology). Define $T : X^* \to c_0$ by

$$T(x^*) = \left(x^*(x_n)\right)_{n\geqslant 1} \in c_0.$$

We define $h : T(X^*) \to \mathbb{K}$ as

$$h\left(T(x^*)\right) = \theta(x^*).$$

We must first show that $\theta$ is well-defined. To this end, suppose $x^* \in X^*$ is such that $Tx^* = 0$, that is, $x^*(x_n) = 0$ for all $n \geqslant 1$. As a result, $Nx^* \in \theta^{-1}(B_{\mathbb{K}}(0,1))$ for each positive integer $N$. As a result,

$|N\theta(x^*)| < 1$ for all positive integers $N \geqslant 1$. Taking $N \to \infty$, we conclude that $\theta(x^*) = 0$. This shows that $h$ is well-defined.

Further, we claim that $h$ is continuous. Indeed, consider $Tx^* = (x^*(x_n))_{n\geqslant 1}$, and suppose

$$\left\| \left(x^*(x_n)\right)_{n\geqslant 1} \right\| = M \geqslant 0.$$

Let $\varepsilon > 0$ and note that

$$\frac{1}{M+\varepsilon} x^* \in \left\{ x^* \in X^* : |x^*(x_n)| < 1 \quad \text{for all } n \geqslant 1 \right\},$$

so that $|\theta(x^*)| < M + \varepsilon$. Since this inequality holds for all $\varepsilon > 0$, it follows that $|\theta(x^*)| \leqslant M$, i.e., $h$ is continuous.

Using the Hahn-Banach Extension Theorem (Theorem 3.2), we may extend $h$ to a continuous linear functional on $c_0$. Recalling that there is an isometric isomorphism $\ell^1 \cong (c_0)^*$, there exists a sequence $(a_n)_{n\geqslant 1} \in \ell^1$ such that

$$h\left((x_n)_{n\geqslant 1}\right) = \sum_{n=1}^{\infty} a_n x_n \qquad \text{for all } (x_n)_{n\geqslant 1} \in c_0.$$

Thus

$$\theta(x^*) = \sum_{n=1}^{\infty} a_n x^*(x_n) = x^*\left(\sum_{n=1}^{\infty} a_n x_n\right),$$

since the sum $\sum_{n=1}^{\infty} a_n x_n =: x_0$ converges in $X$. We have therefore shown that $\theta = \mathrm{ev}_{x_0}$, which is a continuous linear functional with respect to the weak*-topology, as desired. ■

**THEOREM 4.25 (KREIN-ŠMULIAN).** A convex set in $X^*$ is closed in the weak*-topology if and only if it is closed in the $BX$-topology, that is, if and only if its intersection with every positive multiple of the closed unit ball of $X^*$ is weak*-closed.

*Proof.* Let $E \subseteq X^*$ be a convex set. Since the $BX$-topology is finer than the weak*-topology, it suffices to show that closure in the $BX$-topology implies closure in the weak*-topology.

To this end, suppose $E$ is closed in the $BX$-topology, and let $x^* \in X^* \setminus E$. Using the Hahn-Banach separation theorem (Theorem 3.3), there exists a bounded linear functional $\theta : X^* \to \mathbb{K}$ and reals $\gamma_1, \gamma_2$ such that

$$\Re\theta x^* < \gamma_1 < \gamma_2 < \Re\theta y^* \qquad \text{for all } y^* \in E.$$

In view of Theorem 4.24, $\theta$ is continuous in the weak*-topology, so that there is a weak*-open neighborhood of $x^*$ that is disjoint from $E$. This shows that $E$ is weak*-closed, as desired. ■

## §§ The Eberlein-Šmulian Theorem

**LEMMA 4.26 (GOLDSTINE).** Let $X$ be a Banach space. The canonical map $\Phi : X \to X^{**}$ is a homeomorphism from $X$ with the weak topology onto a dense linear subspace of $X^{**}$ with the weak*-topology.

*Proof.* The map $\Phi : X \to X^{**}$ is linear and injective. To show that $\Phi$ is continuous, it suffices to show that it is continuous at 0. Indeed, note that a basis element at 0 in $X^{**}$ is of the form

$$V\left(x_1^*, \ldots, x_n^*; \varepsilon\right) = \left\{ x^{**} \in X^{**} : \left| x^{**}(x_i^*) \right| < \varepsilon \text{ for } 1 \leqslant i \leqslant n \right\}.$$

Then

$$\Phi^{-1}\left(V\left(x_1^*, \ldots, x_n^*; \varepsilon\right)\right) = \left\{ x \in X : \left| x_i^*(x) \right| < \varepsilon \text{ for } 1 \leqslant i \leqslant n \right\} = V\left(x_1^*, \ldots, x_n^*, \varepsilon\right),$$

which is open in $X$. This shows that $\Phi$ is continuous. To see that $\Phi$ is a homeomorphism onto its image, it would suffice to show that $\Phi$ is an open map. Again, since $\Phi$ is linear, it suffices to show that the image of a basic open set at 0 is open in $\Phi(X)$. Indeed, note that

$$\Phi\left(V\left(x_1^*,\ldots,x_n^*;\varepsilon\right)\right) = \Phi(X) \cap V\left(x_1^*,\ldots,x_n^*;\varepsilon\right).$$

This shows that $\Phi$ is a homeomorphism onto its image.

It remains to show that $\Phi(X)$ is weak*-dense in $X^{**}$. Let $B_X$ denote the unit ball of $X$ and $B_{X^{**}}$ the unit ball of $X^{**}$. We shall show that $\Phi(B_X)$ is dense in $X^{**}$ in the weak*-topology. Suppose not, and let $K$ denote the closure of $\Phi(B_X)$ in the weak*-topology, which is a convex set. Choose $x^{**} \in B_{X^{**}} \setminus K$. In view of Theorem 3.3, there exists a bounded linear functional $\Lambda\colon X^{**} \to \mathbb{K}$ and $\gamma_1, \gamma_2 \in \mathbb{R}$ such that

$$\Re\Lambda x^{**} < \gamma_1 < \gamma_2 < \Re\Lambda y^{**}$$

for all $y^{**} \in K$. Since $\Lambda$ is continuous with respect to the weak*-topology. $\Lambda = \mathrm{ev}_{x^*}$ for some $x^* \in X^*$. Further, since the inequality holds for all $y^{**} \in K$, it holds, in particular for all $y^{**} = \Phi(z)$ for some $z \in S_X$, that is,

$$\Re x^{**}(x^*) < \gamma_1 < \gamma_2 < \Re x^*(z)$$

for all $z \in X$ with $\|z\| \leqslant 1$. Taking the infimum over all such $z$, we have

$$\Re\Lambda x^{**} < \gamma_1 < \gamma_2 \leqslant -\|x\|.$$

Therefore, $\|x^{**}(x^*)\| > \|x\|$, which is absurd since $\|x^{**}\| \leqslant 1$. This shows that the image of $B_X$ is dense in $B_{X^{**}}$ in the weak*-topology, thereby completing the proof. ■

**Lemma 4.27.** Let $X$ be a Banach space. Then the closed unit ball $B_{X^*}$ of $X^*$ in the weak*-topology is metrizable if and only if $X$ is separable.

**Lemma 4.28.** Let $f$ be a linear functional on the linear space $X$, and let $\mathscr{F}$ be a total subspace of $X^\vee$. Then the following statements are equivalent:

(1) $f \in \mathscr{F}$,

(2) $f$ is continuous when $X$ is endowed with the $\mathscr{F}$-topology.

(3) $\mathfrak{Z}_f = \{x \in X : f(x) = 0\}$ is closed in the $\mathfrak{Z}_f$-topology.

*Proof.* That (1) and (2) are equivalent and (1) $\implies$ (3) is clear from Theorem 4.6. We show that (3) $\implies$ (1). If $\mathfrak{Z}_f$ is all of $X$, then $f = 0$ and there is nothing to prove. Suppose now that $\mathfrak{Z}_f$ is a proper subspace of $X$ and let $x \in X \setminus \mathfrak{Z}_f$. Since $X$ with the $\mathscr{F}$-topology is a locally convex topological vector space, due to Theorem 3.3 and Theorem 4.6, there is a constant $\gamma \in \mathbb{R}$ and $g \in \mathscr{F}$ such that $\Re g(y) > \gamma$ for all $y \in \mathfrak{Z}_f$. This immediately implies that $g$ is identically zero on $\mathfrak{Z}_f$. In particular, we have shown that $g(y) = 0$ whenever $f(y) = 0$. Due to Lemma 4.1, there is a constant $\alpha \in \mathbb{K}$ such that $g = \alpha f$, so that $f \in \mathscr{F}$, as desired. ■

**Theorem 4.29 (Eberlein-Šmulian).** Let $A$ be a subset of a Banach space $X$. Then the following statements are equivalent:

(1) $A$ is weakly sequentially compact – that is, any sequence in $A$ has a subsequence which converges weakly to an element of $X$.

(2) every countably infinite subset of $A$ has a weak limit point in $X$.

(3) the closure of $A$ in the weak topology is compact (in the weak topology).

*Proof.* First note that each of the three conditions implies the boundedness of $A$ in the norm topology on $X$. Indeed, since each condition implies the boundedness of the set $\{x^*(x) : x \in A\}$ for each $x^* \in X^*$, treating $A$ as a subset of $X^{**}$ through the isometric map $\Phi : X \to X^{**}$ and using the Banach-Steinhaus Theorem, we see that $A$ is a bounded subset of $X$. The implication (3) $\implies$ (2) is a straightforward topological result. We shall show that (2) $\implies$ (1) and (1) $\implies$ (3).

$\boxed{(2) \implies (1)}$ Let $(x_n)_{n=1}^{\infty}$ be a sequence in $A$ and let $X_0 = \overline{\operatorname{Span}}\{x_n : n \geqslant 1\}$, so that $X_0$ is a separable Banach space. By Lemma 4.27, the closed unit ball in $X_0^*$ is metrizable in the weak*-topology, and further, due to Theorem 4.12, the closed unit ball is also compact in this topology so that it is, in particular, separable in the weak*-topology. Since $X_0^*$ is a countable union of multiples of the closed unit ball of $X_0^*$, we see that $X_0^*$ is also separable in the weak*-topology.

Let $H_0$ be a countably dense set in $X_0^*$ with the weak*-topology. Due to Theorem 3.2, each element of $H_0$ can be extended to a continuous linear functional on all of $X$. Fixing one such extension for each element of $H_0$, we get a countable subset $H$ of $X^*$.

We note that $H_0$ is a total set on $X_0$, that is, if $x_0 \in X_0$ with $x^*(x_0) = 0$ for each $x^* \in H_0$, then $x_0 = 0$. Indeed, the evaluation map $\mathrm{ev}_{x_0} : X_0^* \to \mathbb{K}$ is continuous (by construction) when $X_0^*$ is equipped with the weak*-topology. Further, using the density of $H_0$ in $X_0^*$, we conclude that $\mathrm{ev}_{x_0}$ is identically zero, so that $x_0 = 0$.

Using a standard diagonal argument, we may extract a subsequence $(y_m)_{m\geqslant 1}$ of $(x_n)_{n\geqslant 1}$ such that $\lim\limits_{m\to\infty} x^* y_m$ exists for each $x^* \in H$. By condition (2), there is a point $y_0 \in X$ such that every deleted weak neighborhood of $y_0$ contains at least one $y_m$. Since $(y_m)_{m\geqslant 1} \subseteq X_0$ and due to Proposition 4.9, $X_0$ is also weakly closed, we see that $y_0 \in X_0$.

We claim that

$$\lim_{m\to\infty} x^* y_m = x^* y_0 \qquad \text{for every } x^* \in H.$$

Indeed, let $x^* \in H$. By assumption, we already know that the limit $\lim\limits_{m\to\infty} x^* y_m = L$ exists. For any $m \geqslant 1$, we may write

$$\left|x^* y_0 - L\right| \leqslant \left|x^* y_0 - x^* y_m\right| + \left|x^* y_m - L\right|.$$

Let $\varepsilon > 0$ and consider the deleted (weak) neighborhood $U = \left\{y \in X : |x^* y - x^* y_0| < \frac{\varepsilon}{2}\right\} \setminus \{y_0\}$ in $X$. This contains infinitely many of the $y_m$'s. Choose $y_m$ with $m \gg 0$ so that $|x^* y_m - L| < \frac{\varepsilon}{2}$. Then, for such an $m$, we have

$$\left|x^* y_0 - L\right| \leqslant \left|x^* y_0 - x^* y_m\right| + \left|x^* y_m - L\right| < \varepsilon.$$

This inequality holds for all $\varepsilon > 0$ and hence $x^* y_0 = L$, as contended.

Next, we shall show that $\lim\limits_{m\to\infty} x^* y_m = x^* y_0$ for each $x^* \in X^*$. Suppose not, then there exists an $x_0^* \in X^*$, an $\varepsilon > 0$, and a subsequence $(y_{m_k})_{k\geqslant 1}$ such that $\left|x_0^*(y_{m_k} - y_0)\right| > \varepsilon$ for all $k \geqslant 1$. Using condition (2) again to the sequence $(y_{m_k})_{k\geqslant 1}$, there exists an element $y_0' \in X_0$ such that every (deleted) weak neighborhood of $y_0'$ contains at least one of the $y_{m_k}$'s. Arguing as above, we can show that

$$x^* y_0' = \lim_{k\to\infty} x^* y_{m_k} \qquad \text{for all } x^* \in H.$$

Hence, we must have that $x^* y_0 = x^* y_0'$ for all $x^* \in H$, equivalently, $x^*(y_0 - y_0') = 0$ for all $x^* \in H$. Since $H_0$ is total on $X_0$, as we argued earlier, we must have that $y_0 = y_0'$, a fact that contradicts our choice of $x_0$. Thus $(y_m)_{m\geqslant 1}$ converges weakly to $y_0$, as desired.

$\boxed{(1) \implies (3)}$ Let $\overline{A}$ denote the closure of $A$ in the weak*-topology. In view of Lemma 4.26, the map $\Phi : X \to X^{**}$ is a homeomorphism onto a dense linear subspace when $X$ and $X^{**}$ are equipped with the weak and weak*-topologies respectively. So we have $\Phi\left(\overline{A}\right) = \overline{\Phi(A)} \cap \Phi(X)$. Further, $\overline{A}$ is weakly

compact if and only if $\Phi\left(\overline{A}\right)$ is weak*-compact. Recall that as a consequence of Theorem 4.12, a subset of $X^{**}$ is compact in the weak*-topology if and only if it is closed and bounded (in the weak*-topology). Since $\overline{A}$ is bounded, it suffices to show that $\Phi(\overline{A})$ is a weak*-closed subset of $X^{**}$. To this end, we shall show that $\overline{\Phi(A)} \subseteq \Phi(X)$, which is clearly sufficient.

Let $x^{**} \in X^{**}$ be an element in the weak*-closure of $\Phi(A)$. We must show that $x^{**} \in \Phi(X)$. This amounts to showing that ther exists an $x \in X$ such that $x^{**} = \mathrm{ev}_x$ as linear functionals on $X^*$.

**Claim.** If $\{x_1^*, \ldots, x_n^*\}$ is an arbitrary finite subset of $X^*$, then there exists a $z \in \overline{A}$ such that $x^{**}x_i^* = x_i^* z$ for each $1 \leqslant i \leqslant n$.

Consider the following weak*-neighborhood of $x^{**}$:

$$x^{**} + V\left(x_1^*, \ldots, x_n^*; \frac{1}{m}\right),$$

where $m \geqslant 1$ is a positive integer. Since $x^{**}$ is in the closure of $\Phi(A)$, there exists $z_m \in A$ such that $\Phi(z_m)$ is contained in the above neighborhood. That is,

$$\left|x_i^*(z_m) - x^{**}(x_i^*)\right| < \frac{1}{m} \qquad \text{for each } 1 \leqslant i \leqslant n.$$

Since $A$ is sequentially compact, a subsequence of $(z_m)_{m \geqslant 1}$ will converge weakly to an element $z \in \overline{A}$. It is then clear that $x_i^*(z) = x^{**}(x_i^*)$ for each $1 \leqslant i \leqslant n$, thereby proving the claim.

In the remainder of the proof, we shall show that $x^{**} \in \Phi(X)$. Recall that the linear subspace $\Phi(X)$ defines the weak*-topology on $X^*$, and hence, in view of Lemma 4.28, $x^{**} \in \Phi(X)$ if and only if the "zero-set" $Y = \{x^* \in X^* : x^{**}x^* = 0\}$ is weak*-closed in $X^*$. Let $S_{X^*}$ denote the closed unit ball of $X^*$. In view of Theorem 4.25, $Y$ is closed in $X^*$ if and only if $Y \cap rS_{X^*}$ is weak*-closed for every $r > 0$. Since $Y$ is a subspace of $X^*$, this is equivalent to showing that $Y \cap S_{X^*}$ is closed in the weak*-topology on $X^*$.

Thus we must show that if $y_0^*$ is in the weak*-closure of $Y \cap S_{X^*}$, then $y_0^* \in Y \cap S_{X^*}$. In pursuit of this, we choose an arbitrary $\varepsilon > 0$ and construct three sequences $(x_n)_{n \geqslant 1} \subseteq A$, $(y_n^*)_{n \geqslant 0} \subseteq Y \cap S_{X^*}$, and $(z_n)_{n \geqslant 1} \subseteq \overline{A}$ as follows:

From what we remarked earlier, there exists a $z_1 \in \overline{A}$ such that $y_0^*(z_1) = x^{**}(y_0^*)$. Since $z_1$ is in the weak closure of $A$, there exists an $x_1 \in A$ such that $|y_0^*(x_1) - y_0^*(z_1)| < \frac{\varepsilon}{4}$. Since $y_0^*$ is in the weak*-closure of $Y \cap S_{X^*}$, there exists a $y_1^* \in Y \cap S_{X^*}$ such that $|y_1^*(x_1) - y_0^*(x_1)| < \frac{\varepsilon}{4}$.

Inductively, if the elements with subscripts less than $n$ have been chosen, we select $z_n \in \overline{A}$, $x_n \in A$, and $y_n^* \in Y \cap S_{X^*}$ such that

- $y_m^*(z_n) = x^{**}(y_m^*)$ for $0 \leqslant m \leqslant n-1$,
- $|y_m^*(x_n) - y_m^*(z_n)| < \frac{\varepsilon}{4}$ for $0 \leqslant m \leqslant n-1$, and
- $|y_n^*(x_i) - y_0^*(x_i)| < \frac{\varepsilon}{4}$ for $1 \leqslant i \leqslant n$.

The first is achieved by invoking our observation from a preceding paragraph. The second is obtained by considering the weak-neighborhood

$$z_n + V\left(y_1^*, \ldots, y_{n-1}^*; \frac{\varepsilon}{4}\right)$$

of $z_n$, and the third is obtained by considering the weak*-neighborhood

$$y_0^* + V\left(x_1, \ldots, x_n; \frac{\varepsilon}{4}\right).$$

By construction, we have:

- $\|y_m^*\| \leqslant 1$ for all $m \geqslant 1$, and
- $y_m^*(z_n) = 0$ for $1 \leqslant m \leqslant n-1$.

Further, for $1 \leqslant i \leqslant n$, we have

$$|x^{**}(y_0^*) - y_n^*(x_i)| = |y_0^*(z_n) - y_n^*(x_i)| \leqslant |y_0^*(z_n) - y_0^*(x_n)| + |y_0^*(x_n) - y_n^*(x_i)| < \frac{\varepsilon}{2}.$$

Since $(x_n)_{n\geqslant 1}$ is a sequence contained in $A$, and $A$ is weakly sequentially compact, there is a subsequence $(x_{n_k})_{k\geqslant 1}$ which converges weakly to an element $x \in \overline{A}$. To maintain brevity, we shall replace $(x_n)_{n\geqslant 1}$ by this subsequence and tacitly do the same for the sequences $(y_n)_{n\geqslant 1}$ and $(z_n)_{n\geqslant 1}$. Therefore, we may assume that $(x_n)_{n\geqslant 1}$ converges weakly to some $x \in \overline{A}$.

Note that for $n > m$, we have

$$|y_m^*(x_n)| = |y_m^*(x_n) - y_m^*(z_n)| < \frac{\varepsilon}{4},$$

so that in the weak limit, we have $y_m^*(x)| \leqslant \frac{\varepsilon}{4}$. Due to Proposition 4.9, there is a convex combination $w = \sum_{i=1}^{N} a_i x_i$ such that $|x - w| < \frac{\varepsilon}{4}$. Therefore,

$$|x^{**}(y_0^*) - y_N^*(w)| \leqslant \sum_{i=1}^{N} a_i |x^{**}(y_0^*) - y_N^*(x_i)| < \frac{\varepsilon}{2}.$$

It follows that

$$|x^{**}(y_0^*)| \leqslant |x^{**}(y_0^*) - y_N^*(w)| + |y_N^*(w) - y_N^*(x)| + |y_N(x)|.$$

Note that the first term on the right hand side is smaller than $\frac{\varepsilon}{2}$ as argued above. The second is less than $\frac{\varepsilon}{4}$ since $\|y_N^*\| \leqslant 1$, and the final term is at most $\frac{\varepsilon}{4}$ as remarked above. In particular, the right hand side is smaller than $\varepsilon$. Since this choice of $\varepsilon > 0$ was arbitrary, it follows that $x^{**}(y_0^*) = 0$, in other words, $y_0^* \in Y$, and hence, $y_0^* \in Y \cap S_{X^*}$, as desired.

This shows that $Y$ is weak*-closed, so that $x^{**} \in \Phi(X)$, and the proof is now complete. ■

## §5 Duality in Banach Spaces

Throughout this section, for a normed linear space $X$, we use $x^*$ to denote the elements of its dual $X^*$. Further, there is a natural pairing

$$\langle \cdot, \cdot \rangle : X \times X^* \to \mathbb{K} \qquad \langle x, x^* \rangle = x^*(x).$$

**Definition 5.1.** Suppose $X$ is a Banach space, $M$ is a subspace of $X$, and $N$ is a subspace of $X^*$. Their *annihilators* $M^\perp$ and ${}^\perp N$ are defined as follows

$$M^\perp = \{x^* \in X^* : \langle x, x^* \rangle = 0 \text{ for all } x \in M\}$$
$${}^\perp N = \{x \in X : \langle x, x^* \rangle = 0 \text{ for all } x^* \in N\}.$$

Obviously, $M^\perp$ is weak*-closed in $X^*$ and ${}^\perp N$ is norm-closed in $X$.

**Theorem 5.2.** Let $X$ be a normed linear space, $M$ a subspace of $X$ and $N$ a subspace of $X^*$.

(a) ${}^\perp(M^\perp)$ is the norm-closure of $M$ in $X$.

(b) $({}^{\perp}N)^{\perp}$ is the weak*-closure of $N$ in $X^*$.

*Proof.* Obviously $M \subseteq {}^{\perp}(M^{\perp})$, and the latter is norm closed in $X$, and hence contains the norm closure of $M$. On the other hand, if $x$ is not in the norm closure of $M$, then due to the Hahn-Banach theorem, there is an $x^* \in X^*$ such that $\langle x, x^* \rangle \neq 0$ but $x^*$ vanishes on $M$. Hence, $x \notin {}^{\perp}(M^{\perp})$.

Simiarly, $N$ is contained in $({}^{\perp}N)^{\perp}$, which is weak*-closed, therefore contains the weak*-closure of $N$. On the other hand, if $x^*$ is not in the weak*-closure of $N$, using the fact that $X^*$ is locally convex under the weak*-topology, there is a continuous linear functional $\Lambda : X^* \to \mathbb{K}$ (w.r.t. the weak*-topology) that vanishes on $N$ but not $x^*$. But $\Lambda = \mathrm{ev}_x$ for some $x \in X$ and since $\Lambda$ vanishes on $N$, we have that $x \in {}^{\perp}N$. Therefore, $x \notin ({}^{\perp}N)^{\perp}$. ■

**THEOREM 5.3.** Suppose $X$ and $Y$ are normed linear spaces, and $T \in \mathscr{B}(X,Y)$. Then

$$\mathscr{N}(T^*) = \mathscr{R}(T)^{\perp} \qquad \text{and} \qquad \mathscr{N}(T) = {}^{\perp}\mathscr{R}(T^*).$$

*Proof.* The proof is quite straightforward.

$$\begin{aligned} y^* \in \mathscr{N}^*(T) &\iff T^* y^* = 0 \iff \langle x, T^* y^* \rangle \ \forall x \in X \\ &\iff \langle Tx, y^* \rangle = 0 \ \forall x \in X \iff y^* \in \mathscr{R}(T)^{\perp}. \end{aligned}$$

Similarly,

$$\begin{aligned} x \in \mathscr{N}(T) &\iff Tx = 0 \iff \langle Tx, y^* \rangle = 0 \ \forall y^* \in X^* \\ &\iff \langle x, T^* y^* \rangle = 0 \ \forall y^* \in X^* \iff x \in {}^{\perp}\mathscr{R}(T^*). \end{aligned}$$

This completes the proof. ■

**COROLLARY.** Let $T \in \mathscr{B}(X,Y)$ where $X$ and $Y$ are normed linear spaces.

(a) $\mathscr{N}(T^*)$ is weak*-closed in $Y^*$.

(b) $\mathscr{R}(T)$ is dense in $Y$ if and only if $T^*$ is injective.

(c) $T$ is injective if and only if $\mathscr{R}(T^*)$ is weak*-dense in $X^*$.

*Proof.* All three follow immediately from Hahn-Banach and the preceding result. ■

**THEOREM 5.4.** Let $U$ and $V$ be the open unit balls in the Banach spaces $X$ and $Y$ respectively. If $T \in \mathscr{B}(X,Y)$, the following are equivalent:

(a) There is a $\delta > 0$ such that $\|T^* y^*\| \geqslant \delta \|y^*\|$ for every $y^* \in Y^*$.

(b) $\overline{T(U)} \supseteq \delta V$.

(c) $T(U) \supseteq \delta V$.

(d) $T(X) = Y$.

*Proof.* Suppose (a) holds and chooe $y_0 \notin \overline{T(U)}$. Using the Hahn-Banach separation theorem, choose a $y^* \in Y^*$ such that $|\langle y, y^* \rangle| \leqslant 1$ for every $y \in \overline{T(U)}$, but $|\langle y_0, y^* \rangle| > 1$. Thus, if $x \in U$, then we have

$$|\langle x, T^* y^* \rangle| = |\langle Tx, y^* \rangle| \leqslant 1.$$

Thus $\|T^*y^*\| \leqslant 1$, whence it follows from (a) that

$$\|y_0\| \geqslant \|y_0\|\|T^*y^*\| \geqslant \delta\|y_0\|\|y^*\| \geqslant \delta|\langle y_0, y^*\rangle| > \delta.$$

Consequently, if $\|y\| \leqslant \delta$, then $y \in \overline{T(U)}$, as desired.

Next, suppose (b) holds. Replacing $T$ by $\delta^{-1}T$, we may suppose that $\delta = 1$, that is, $V \subseteq \overline{T(U)}$, whence $\overline{V} \subseteq \overline{T(U)}$. If $y \in Y$ is non-zero, and $\varepsilon > 0$, then $y/\|y\| \in \overline{V}$, and we can find an $x_0 \in U$ such that $\|Tx_0 - y/\|y\|\| < \varepsilon/\|y\|$, therefore, there is an $x \in X$ with $\|x\| \leqslant \|y\|$ such that $\|Tx - y\| < \varepsilon$.

We shall now show that $V \subseteq T(U)$. Pick some $y \in V$ and set $y_1 = y$. Choose a sequence $(\varepsilon_n)$ of positive reals such that

$$\sum_{n=1}^{\infty} \varepsilon_n < 1 - \|y_1\|.$$

We shall now define two sequences $(x_n)$ and $(y_n)$. Let $n \geqslant 1$ and suppose $y_n$ has been chosen. Then there is an $x_n \in X$ such that $\|x_n\| \leqslant \|y_n\|$ and $\|y_n - Tx_n\| < \varepsilon_n$. Set

$$y_{n+1} = y_n - Tx_n.$$

Note that for $n \geqslant 1$,

$$\|x_{n+1}\| \leqslant \|y_{n+1}\| < \varepsilon_n,$$

according to our construction. Hence, the sequence $(x_n)$ is absolutely summable and since we are in a Banach space, it is summable. It follows that

$$\|x\| := \left\|\sum_{n=1}^{\infty} x_n\right\| \leqslant \sum_{n=1}^{\infty} \|x_n\| < \|x_1\| + \sum_{n=1}^{\infty} \varepsilon_n < 1,$$

since $\|x_1\| \leqslant \|y_1\|$. Consequently, $x \in U$, and

$$Tx = \lim_{N\to\infty} \sum_{n=1}^{N} Tx_n = \lim_{N\to\infty} \sum_{n=1}^{\infty} y_n - y_{n+1} = y_1 = y,$$

as desired.

If (c) holds, then using the fact that $V$ is absorbing in $Y$, it is immediate that $T$ is surjective.

Finally, suppose (d) holds. Due to the open mapping theorem, there is a $\delta > 0$ such that $\delta V \subseteq T(U)$. Hence

$$\begin{aligned}
\|T^*y^*\| &= \sup\left\{|\langle x, T^*y^*\rangle| : x \in U\right\} \\
&= \sup\left\{|\langle Tx, y^*\rangle| : x \in U\right\} \\
&\geqslant \sup\left\{|\langle y, y^*\rangle| : y \in \delta V\right\} \\
&= \delta \sup\left\{|\langle y, y^*\rangle| : y \in V\right\} = \delta\|y^*\|.
\end{aligned}$$

This completes the proof. ■

**THEOREM 5.5 (CLOSED RANGE THEOREM).** If $X$ and $Y$ are Banach spaces and $T \in \mathscr{B}(X, Y)$, then the following are equivalent:

(a) $\mathscr{R}(T)$ is norm-closed in $Y$.

(b) $\mathscr{R}(T^*)$ is weak*-closed in $X^*$.

(c) $\mathscr{R}(T^*)$ is norm-closed in $X^*$.

*Proof.* Suppose first that (a) holds. We shall show that $\mathscr{R}(T^*)$ is its own weak*-closure. Recall that the weak*-closure of $\mathscr{R}(T^*)$ is given by $({}^{\perp}\mathscr{R}(T^*))^{\perp} = \mathscr{N}(T)^{\perp}$. Therefore, it suffices to show that $\mathscr{N}(T)^{\perp} \subseteq \mathscr{R}(T^*)$.

Pick $x^* \in \mathscr{N}(T)^{\perp}$ and define a linear functional $\Lambda$ on $\mathscr{R}(T)$ by

$$\Lambda(Tx) = \langle x, x^* \rangle \qquad \forall\ x \in X.$$

This functional is well-defined, for if $Tx = Tx'$, then $x - x' \in \mathscr{N}(T)$ and thus $\langle x - x', x^* \rangle = 0$. Next, since $\mathscr{R}(T)$ is closed in $Y$, it is a Banach space and hence, the open mapping theorem applies, consequently, there is a constant $K > 0$ such that for each $y \in \mathscr{R}(T)$, there is an $x \in X$ with $Tx = y$ and $\|x\| \leqslant K\|y\|$. Hence,

$$|\Lambda y| = |\Lambda(Tx)| = |\langle x, x^* \rangle| \leqslant \|x\|\|x^*\| \leqslant K\|y\|\|x^*\|,$$

i.e., $\Lambda$ is continuous. This can then be extended to a linear functional $y^* \in Y^*$. Hence, for all $x \in X$, we have

$$\langle Tx, y^* \rangle = \Lambda(Tx) = \langle x, x^* \rangle.$$

Thus $x^* = T^* y^*$, as desired.

Obviously, if (b) holds, then (c) does, since the norm topology on $X^*$ is finer than the weak*-topology.

Suppose now that (c) holds. Let $Z$ denote the norm-closure of $\mathscr{R}(T)$ in $Y$ and let $S$ denote the corestriction of $T$ to $Z$. Due to 19 5 (b), since $\mathscr{R}(S)$ is dense in $Z$, $S^* : Z^* \to X^*$ is injective.

If $z^* \in Z^*$, we can extend this to some $y^* \in Y^*$ using Hahn-Banach. Then, for every $x \in X$, we have

$$\langle x, T^* y^* \rangle = \langle Tx, y^* \rangle = \langle Sx, z^* \rangle = \langle x, S^* z^* \rangle.$$

Hence, $S^* z^* = T^* y^*$, consequently, $\mathscr{R}(S^*) = \mathscr{R}(T^*)$, is norm-closed due to (c), and hence, complete. It follows from the open mapping theorem that $S^* : Z^* \to \mathscr{R}(S^*)$ is an isomorphism, owing to it being continuous and bijective between Banach spaces. Hence, there is a constant $c > 0$ such that

$$c\|z^*\| \leqslant \|S^* z^*\| \qquad \forall\ z^* \in Z^*.$$

Due to Theorem 5.4, $S : X \to Z$ is surjective. But since $\mathscr{R}(T) = \mathscr{R}(S)$, we have that $\mathscr{R}(T) = Z$ is a closed subspace of $Y$, thereby completing the proof. ■

## §6 Compact Operators

**Definition 6.1.** A linear map $T : X \to Y$ between Banach spaces is said to be *compact* if $T(U)$ is relatively compact in $Y$ where $U$ is the unit ball in $X$.

The following proposition is immediate from the equivalence of compactness and sequential compactness in metric spaces.

**Proposition 6.2.** $T$ is compact if and only if every bounded sequence $(x_n)$ in $X$ contains a subsequence $(x_{n_k})$ such that $(Tx_{n_k})$ converges in $Y$.

**Definition 6.3.** The *spectrum* $\sigma(T)$ of an operator $T \in \mathscr{B}(X)$ is the set of all scalars $\lambda$ such that $T - \lambda I$ is not invertible.

**Theorem 6.4.** Let $X$ and $Y$ be Banach spaces.

(a) If $T \in \mathscr{B}(X, Y)$ and $\dim \mathscr{R}(T) < \infty$, then $T$ is compact.

(b) If $T \in \mathscr{B}(X,Y)$, $T$ is compact, and $\mathscr{R}(T)$ is closed, then $\dim \mathscr{R}(T) < \infty$.

(c) The compact operators form a closed subspace of $\mathscr{B}(X,Y)$ in its norm-topology.

(d) If $T \in \mathscr{B}(X)$, $T$ is compact, and $\lambda \neq 0$ is a scalar, then $\dim \mathscr{N}(T - \lambda I) < \infty$.

(e) If $\dim X = \infty$, $T \in \mathscr{B}(X)$, and $T$ is compact, then $0 \in \sigma(T)$.

(f) If $S, T \in \mathscr{B}(X)$, and $T$ is compact, then so are $ST$ and $TS$.

*Proof.* (a) Let $U$ denote the unit ball of $X$. Then $T(U)$ is a bounded subset of $\mathscr{R}(T)$ and since the latter is closed in $Y$, $\overline{T(U)}$ is a closed and bounded subset of $\mathscr{R}(T)$, consequently, is compact.

(b) Since $\mathscr{R}(T)$ is closed in $Y$, it is complete, i.e., a Banach space. Due to the open mapping theorem, $T(U)$ is open in $\mathscr{R}(T)$ with compact closure, whence $\mathscr{R}(T)$ is locally compact, and hence, finite dimensional.

(c) Let $T_n \to T$ in $\mathscr{B}(X,Y)$ where each $T_n$ is a compact operator. We shall show that $T(U)$ is totally bounded in $Y$. Let $\varepsilon > 0$ and choose an $N$ such that $\|T - T_N\| < \varepsilon/3$. Note that $T_N(U)$ is totally bounded in $Y$, and hence, there are $x_1, \ldots, x_n \in U$ such that

$$T_N(U) \subseteq \bigcup_{i=1}^{n} B_Y(T_N x_i, \varepsilon/3).$$

Now, for any $y \in U$, there is an index $1 \leqslant i \leqslant n$ such that $T_N y \in B(T_N x_i, \varepsilon/3)$. As a result,

$$\|Ty - Tx_i\| \leqslant \|Ty - T_N y\| + \|T_N y - T_N x_i\| + \|T_N x_i - Tx_i\| < \varepsilon.$$

Hence,

$$T(U) \subseteq \bigcup_{i=1}^{n} B_Y(Tx_i, \varepsilon),$$

and the conclusion follows.

(d) Let $Y = \mathscr{N}(T - \lambda I)$. Then note that $T$ acts on $Y$ by $y \mapsto \lambda y$. Further, since $T$ is compact and $Y$ is closed in $X$, the restriction of $T$ to $Y$ is compact and hence, $Y$ must be finite-dimensional.

(e) If $0 \notin \sigma(T)$, then $T$ is invertible, whence $\mathscr{R}(T)$ is closed but $\dim \mathscr{R}(T) = \infty$, a contradiction.

(f) This follows from Proposition 6.2. ■

**THEOREM 6.5.** Suppose $X$ and $Y$ are Banach spaces and $T \in \mathscr{B}(X,Y)$. Then $T$ is compact if and only if $T^* \in \mathscr{B}(Y^*, X^*)$ is compact.

*Proof.* Suppose first that $T$ is compact and let $\{y_n^*\}$ be a sequence in the unit ball of $Y^*$. We shall show that $T^* y^* = y^* \circ T$ admits a convergent subsequence in $X^*$. Let $K = \overline{T(U)} \subseteq Y$, which, according to our assumption is compact in $Y$. Note that the collection $\{y_n^*\}$ is equicontinuous and pointwise bounded on $K$. Due to the Arzelá-Ascoli Theorem, there is a subsequence $\{y_{n_k}^*\}$ that converges uniformly on $K$.

We contend that $\{T^* y_{n_k}^*\}$ converges in the operator norm. Indeed, for any $x \in U$,

$$|(T^* y_{n_k}^*(x) - T^* y_{n_l}^*(x)| = |y_{n_k}^*(Tx) - y_{n_l}^*(Tx)|,$$

and since $Tx \in K$, the conclusion follows.

Conversely, suppose $T^*$ is compact. Consider the natural isometric embeddings $\Phi_X : X \to X^{**}$ and $\Phi_Y : Y \to Y^{**}$, which fit into a commutative diagram

$$\begin{tikzcd} X \arrow[r, "X"] \arrow[d, "\Phi_X"'] & Y \arrow[d, "\Phi_Y"] \\ X^{**} \arrow[r, "T^{**}"'] & Y^{**}. \end{tikzcd} \tag{2}$$

Due to the first part of the proof, $T^{**}$ is compact. Thus, $T^{**}(U^{**})$ is totally bounded in $Y^{**}$. Next, $\Phi_X(U)$ is contained in $U^{**}$ and hence, $T^{**}\Phi_X(U) = \Phi_Y T(U)$ is totally bounded in $Y^{**}$. Since $\Phi_Y$ is an isometry, it follows that $T(U)$ is totally bounded in $Y$, thereby completing the proof. ■

**Definition 6.6.** A closed subspace $M$ of a topological vector space $X$ is said to be *complemented* if there exists a closed subspace $N$ of $X$ such that

$$X = M + N \quad \text{and} \quad M \cap N = \{0\}.$$

In this case, $X$ is said to be the *direct sum* of $M$ and $N$, denoted by $X = M \oplus N$.

**Lemma 6.7.** Let $M$ be a closed subspace of a topological vector space $X$.

(a) If $X$ is locally convex and $\dim M < \infty$, then $M$ is complemented in $X$.

(b) If $\dim(X/M) < \infty$, then $M$ is complemented in $X$.

*Proof.* (a) Let $\{e_1, \dots, e_n\}$ be a basis for $M$. Every $x \in M$ has a unique representation

$$x = \alpha_1(x)e_1 + \cdots + \alpha_n(x)e_n.$$

Note that $\alpha_i(e_j) = 0$ whenever $i \neq j$. Due to the Hahn-Banach Theorem, each $\alpha_i$ can be extended to a continuous linear functional on $X$. Let $N = \bigcap_{i=1}^{n} \mathscr{N}(\alpha_i)$. It is not hard to argue that $X = M \oplus N$.

(b) Let $\pi : X \to X/M$ be the quotient map, and let $\{e_1, \dots, e_n\}$ be a basis for $X/M$. Lift this to $\{x_1, \dots, x_n\}$ in $X$ and let $N$ be the vector subspace they span. Again, it is not hard to argue that $X = M \oplus N$. ■

**Theorem 6.8.** Let $X$ be a Banach space, $T \in \mathscr{B}(X)$ a compact operator, and $\lambda \neq 0$. Then $T - \lambda I$ has closed range.

*Proof.* Let $N = \mathscr{N}(T - \lambda I)$, which is a closed subspace of $X$. Due to Lemma 6.7, admits a complement, say $M$. Let $S : M \to X$ be given by $x \mapsto Tx - \lambda x$, which is a bounded linear operator. Since $\mathscr{R}(S) = \mathscr{R}(T - \lambda I)$, it suffices to show that the former is closed.

To this end, we first show that there is a constant $\beta > 0$ such that $\|Sx\| \geqslant \beta\|x\|$ for all $x \in M$, which is equivalent to

$$\beta = \inf_{\substack{\|x\|=1 \\ x \in M}} \|Sx\| > 0.$$

Suppose not. Then, there is a sequence $x_n \in M$ with $\|x_n\| = 1$, such that $Sx_n \to 0$ as $n \to \infty$. Since $T : X \to X$ is compact, its restriction to $M$ is also compact, whence, there is a subsequence $(x_{n_k})$ such that $Tx_{n_k} \to x_0$ for some $x_0 \in X$. Replace $x_n$ with this subsequence. Then, $Tx_n - \lambda x_n \to 0$ and hence, $\lambda x_n \to x_0$. As a result,

$$Sx_0 = \lim_{n\to\infty} S(\lambda x_n) = \lambda \lim_{n\to\infty} Sx_n = 0.$$

But since $S$ is injective, $x_0 = 0$. This is absurd, since $\|x_0\| = \lim_{n\to\infty} \|\lambda x_n\| = |\lambda| > 0$. It follows that $\beta > 0$.

Finally, we show that $\mathscr{R}(S)$ is closed in $X$. Indeed, suppose $y \in \overline{\mathscr{R}(S)}$; then there is a sequence $(x_n)$ in $M$ such that $Sx_n \to y$, that is $(Sx_n)$ is Cauchy. But since

$$\beta\|x_n - x_m\| \leqslant \|Sx_n - Sx_m\|,$$

so is $(x_n)$. Hence, $x_n \to x_0$ for some $x_0 \in M$; and $Sx_0 = y$. This completes the proof. ■

**THEOREM 6.9 (SPECTRUM OF A COMPACT OPERATOR).** Let $X$ be a Banach space and $T \in \mathscr{B}(X)$ a compact operator.

(a) Every $0 \neq \lambda \in \sigma(T)$ is an eigenvalue of $T$.

(b) For every $\lambda \neq 0$, the increasing chain of subspaces

$$\mathscr{N}(T - \lambda I) \subseteq \mathscr{N}((T - \lambda I)^2) \subseteq \cdots$$

eventually stabilizes. Further, a these subspaces are finite dimensional.

(c) For every $r > 0$, the set

$$\{\lambda \in \sigma(T) \colon |\lambda| > r\}$$

is finite.

(d) As a consequence, $\sigma(T)$ is countable and the only possible limit point of $\sigma(T)$ is 0.

*Proof.* Suppose $\dim X = \infty$, for if $\dim X < \infty$, then all the above statements are trivial as there are only finitely many eigenvalues.

(a) Suppose $0 \neq \lambda \in \sigma(T)$ is not an eigenvalue of $T$, then $T - \lambda I$ is injective, but not surjective, else, due to the open mapping theorem, it would be invertible. Define

$$Y_n = (T - \lambda I)^n(X).$$

Obviously, $Y_{n+1} \subseteq Y_n$ for all $n \geqslant 1$. Further, since the restriction of $T$ to each of these subspaces is compact, due to Theorem 6.4 (d), each $Y_n$ is infinite-dimensional and all inclusions are strict.

For each $n \geqslant 1$, using the Riesz Lemma, choose $y_n \in Y_n \setminus Y_{n+1}$ such that $\|y_n\| = 1$ and

$$\operatorname{dist}(y_n, Y_{n+1}) > \frac{1}{2}.$$

Since $T$ is compact and $(x_n)$ is bounded, the sequence $(Tx_n)$ must admit a convergent subsequence. But for $n < m$, we have

$$\|Tx_n - Tx_m\| = \|(T - \lambda I)x_n + \lambda x_n - (T - \lambda I)x_m - \lambda x_m\|,$$

and since $(T - \lambda I)x_n - (T - \lambda I)x_m - \lambda x_m \in Y_{n+1}$, we conclude that $\|Tx_n - Tx_m\| > \lambda/2$, a contradiction.

(b) If $\lambda$ is not an eigenvalue, then each $\mathscr{N}((T-\lambda I)^n)$ is the trivial subspace and there is nothing to prove. Suppose now that $\lambda$ is an eigenvalue of $T$ and set $Y_n = \mathscr{N}((T-\lambda I)^n)$. Obviously $Y_1 \subseteq Y_2 \subseteq \cdots$. Further, $(T-\lambda I)^n = S + (-\lambda)^n I$ where $S$ is some compact operator and hence, $\dim Y_n < \infty$. Next, note that if $Y_n = Y_{n+1}$ for some $n \geqslant 1$, then $Y_n = Y_{n+1} = Y_{n+2} = \cdots$.

Suppose now that $Y_n \subsetneq Y_{n+1}$ for every $n \geqslant 1$. Again, using the Riesz Lemma, choose $y_{n+1} \in Y_{n+1} \setminus Y_n$ such that $\|y_{n+1}\| = 1$ and

$$\operatorname{dist}(y_{n+1}, Y_n) > \frac{1}{2}.$$

Again, since $(y_n)$ is bounded and $T$ is compact, the sequence $(Ty_n)$ must admit a convergent subsequence. But for $2 \leqslant n < m$, we have

$$\|Ty_n - Ty_m\| = \|(T-\lambda I)y_n + \lambda y_n - (T-\lambda I)y_m - \lambda y_m\|,$$

and since $(T-\lambda I)y_n - (T-\lambda I)y_m + \lambda y_n \in Y_{m-1}$, it follows that $\|Ty_n - Tx_m\| > \lambda/2$, a contradiction.

(c) Suppose there is an $r > 0$ such that the set $\{\lambda \in \sigma(T) \colon |\lambda| > r\}$ is infinite. Choose a countable subset $\{\lambda_1, \lambda_2, \ldots\}$ with corresponding eigenvectors $\{x_1, x_2, \ldots\}$. Let $Y_n = \operatorname{span}\{x_1, \ldots, x_n\}$; when then form a strictly increasing chain of closed subspaces.

First, we contend that for $n \geqslant 2$, $(T-\lambda_n I)(Y_n) \subseteq Y_{n-1}$. Indeed, any element of $Y_n$ can be written uniquely as

$$Y_n \ni y = \alpha_1 x_1 + \cdots + \alpha_n x_n.$$

Then, $(T-\lambda_n I)y = \alpha_1(T-\lambda_n I)x_1 + \cdots + \alpha_{n-1}(T-\lambda_n I)x_{n-1}$. And for $1 \leqslant i \leqslant n-1$, we have

$$(T-\lambda_i I)(T-\lambda_n)x_i = (T-\lambda_n I)(T-\lambda_i I)x_i = 0,$$

whence $(T-\lambda_n)x_i \in Y_i$.

Next, using the Riesz Lemma, for $n \geqslant 2$, choose $y_n \in Y_n \setminus Y_{n-1}$ such that $\|y_n\| = 1$ and

$$\operatorname{dist}(y_n, Y_{n-1}) > \frac{1}{2}.$$

Since $(y_n)$ is bounded and $T$ is compact, the sequence $(Ty_n)$ admits a convergent subsequence. But for $2 \leqslant n < m$, we have

$$\|Ty_n - Ty_m\| = \|(T-\lambda_n I)y_n + \lambda_n y_n - (T-\lambda_m I)y_m - \lambda_m y_m\|,$$

and since

$$(T-\lambda_n I)y_n + \lambda_n y_n - (T-\lambda_m I)y_m \in Y_{m-1},$$

we get that $\|Ty_n - Ty_m\| > |\lambda_m|/2 > r/2$, a contradiction.

(d) Note that

$$\sigma(T) = \{0\} \cup \bigcup_{n \geqslant 1} \left\{ \lambda \in \sigma(T) \colon |\lambda| > \frac{1}{n} \right\},$$

and being a countable union of finite sets, $\sigma(T)$ is countable. Next, suppose $0 \neq \mu \in \mathbb{K}$ is a limit point of $\sigma(T)$. There is an $\varepsilon > 0$ such that $|\mu| > \varepsilon$. But since the set of eigenvalues in $\mathbb{K} \setminus \overline{B}(0, \varepsilon)$ is finite, $\mu$ cannot be their limit point. This completes the proof. ■

## §§ Examples

**THEOREM 6.10 (MINKOWSKI'S INTEGRAL INEQUALITY).** Let $(X, \mathfrak{M}, \mu)$ and $(Y, \mathfrak{N}, \lambda)$ be positive measure spaces. If $f : X \times Y \to \mathbb{R}$ is non-negative and measurable with respect to the product measure, then for $1 \leqslant p < \infty$,

$$\left\{\int_X \left(\int_Y f(x,y)\, d\lambda(y)\right)^p \, d\mu(x)\right\}^{\frac{1}{p}} \leqslant \int_Y \left(\int_X f(x,y)^p \, d\mu(x)\right)^{\frac{1}{p}} \, d\lambda(y)$$

*Proof.* Since $p = 1$ is just Fubini, we assume $p > 1$ and let $q$ be the conjugate exponent to $p$. Let $H : X \to \mathbb{R}$ be defined as

$$H(x) = \int_Y f(x,y)\, d\lambda(y),$$

which is a measurable function on $X$ due to Fubini. We now have the series of inequalities

$$\begin{aligned}
\|H\|_p^p &= \int_X \int_Y f(x,y) H(x)^{p-1} \, d\lambda(y) d\mu(x) \\
&= \int_Y \int_X f(x,y) H(x)^{p-1} \, d\mu(x) d\lambda(y) \\
&\leqslant \int_Y \left(\int_X f(x,y)^p d\mu(x)\right)^{\frac{1}{p}} \left(\int_X H(x)^{pq-q}\right)^{\frac{1}{q}} \, d\lambda(y) \\
&= \int_Y \left(\int_X f(x,y)^p \, d\mu(x)\right)^{\frac{1}{p}} \|H\|_p^{\frac{p}{q}} \, d\lambda(y)
\end{aligned}$$

and hence

$$\|H\|_p \leqslant \int_X \left(\int_X f(x,y)^p \, d\mu(x)\right)^{\frac{1}{p}} \, d\lambda(y),$$

thereby completing the proof. ■

**THEOREM 6.11.** Let $1 < p < \infty$ and define the *Hardy operator* $H : L^p(0,\infty) \to L^p(0,\infty)$ as

$$Hf(x) = \frac{1}{x} \int_0^x f(t)\, dt.$$

Then, $H$ is a non-compact operator with operator norm

$$\|H\| = \frac{p}{p-1}.$$

*Proof.* For operator norm, take $x^{-1/p} \chi_{[0,N]}$ and let $N \to \infty$. ■

# §7 REFLEXIVE SPACES

**DEFINITION 7.1.** A normed linear space $X$ is said to be *reflexive* if the natural embedding $\Phi : X \to X^{**}$ is surjective.

**PROPOSITION 7.2.** Let $X$ be a normed linear space. The natural embedding $\Phi : X \to X^{**}$ is a topological imbedding when $X$ is given the weak topology and $X^*$ is given the weak*-topology.

*Proof.* ■

**THEOREM 7.3 (KAKUTANI).** A Banach space $X$ is reflexive if and only if its norm-closed unit ball is weakly compact.

*Proof.* Let $B, B^{**}$ denote the norm-closed unit balls of $X$ and $X^{**}$ respectively. If $X$ were reflexive, then the natural embedding $\Phi : X \to X^{**}$ is surjective. Due to the preceding result, $\Phi$ is a homeomorhpism when $X$ is given the weak topology and $X^{**}$ is given the weak*-topology. Since $B^{**}$ is compact in the weak*-topology, and $\Phi$ is an isometry, we see that $B$ must be compact in the weak topology.

Conversely, suppose $B$ is compact in the weak topology. Again, due to the preceding proposition, $\Phi(B)$ is compact and convex in the weak*-topology and $\Phi(B) \subseteq B^{**}$. If $X$ were not reflexive, then $\Phi(B) \subsetneq B^{**}$. Choose $x^{**} \in B^{**} \setminus \Phi(B)$. Due to the Hahn-Banach Separation Theorem, there is a linear functional $\Lambda : X^{**} \to \mathbb{K}$ that is continuous with respect to the weak*-topology on $X^*$ and there are $\gamma_1, \gamma_2 \in \mathbb{R}$ such that

$$\Re\Lambda(x^{**}) < \gamma_1 < \gamma_2 < \Re\Lambda(y) \quad \forall y \in \Phi(B).$$

Note that there is some $0 \neq x^* \in X^*$ such that $\Lambda = \mathrm{ev}_{x^*}$, and hence,

$$\Re x^{**}(x^*) < \gamma_1 < \gamma_2 \leqslant \inf_{y \in \Phi(B)} \Re y(x^*) = \inf_{x \in B} \Re x^*(x).$$

The rightmost quantity is precisely $-\|x^*\|$. Thus $\Re x^{**}(x^*) < -\|x^*\|$, in particular, $|x^{**}(x^*)| > \|x^*\|$, whence $\|x^{**}\| > 1$, a contradiction, since we chose it inside $B^{**}$. This completes the proof. ■

**COROLLARY.** Every closed, bounded convex subset of a reflexive Banach space is weakly compact.

*Proof.* This follows from the fact that a convex closed subset of an LCTVS is also weakly closed. ■

## §8 HILBERT SPACES

**DEFINITION 8.1.** An *inner product space* is a $\mathbb{K}$-vector space $H$ together with a function $(\cdot,\cdot) : H \times H \to \mathbb{K}$ such that

(i) $(x, y) = \overline{(y, x)}$,

(ii) $(x + y, z) = (x, z) + (y, z)$,

(iii) $(\alpha x, y) = \alpha(x, y)$,

(iv) $(x, x) \geqslant 0$, and $(x, x) = 0$ if and only if $x = 0$,

for all $x, y, z \in H$ and $\alpha \in \mathbb{K}$.

Obviously, $\|x\| := \sqrt{(x, x)}$ defines a norm on $H$. If $H$ is complete with respect to this norm, then $H$ is said to be a *Hilbert space*.

**PROPOSITION 8.2.** Let $H$ be an inner product space and $x, y \in H$. Then,

$$|(x, y)| \leqslant \|x\|\|y\| \quad \text{and} \quad \|x + y\| \leqslant \|x\| + \|y\|.$$

*Proof.* For every $\lambda \in \mathbb{K}$, we have

$$0 \leqslant (x + \lambda y, x + \lambda y) = |\lambda|^2\|y\|^2 + \|x\|^2 + 2\Re(x, \lambda y).$$

For every $\alpha \in \mathbb{R}$, we can choose $\lambda \in \mathbb{K}$ such that $|\lambda| = |\alpha|$ and $\Re(x, \lambda y) = \alpha|(x, y)|$. Thus,

$$\alpha^2\|y\|^2 + 2\alpha(x, y) + \|x\|^2 \geqslant 0$$

for every $\alpha \in \mathbb{R}$. Thus,

$$4|(x,y)|^2 \leqslant 4\|x\|^2\|y\|^2 \implies |(x,y)| \leqslant \|x\|\|y\|. \tag{3}$$

Finally, note that

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2 + 2\Re(x,y) \leqslant \|x\|^2 + \|y\|^2 + 2|(x,y)| \leqslant (\|x\| + \|y\|)^2,$$

thereby completing the proof. ■

**THEOREM 8.3.** Let $H$ be a Hilbert space. Every nonempty closed convex $E \subseteq H$ contains a unique $x$ of minimal norm.

*Proof.* Let

$$d = \inf\{\|x\| : x \in E\}.$$

Choose a sequence $(x_n)$ in $E$ such that $\|x_n\| \to d$ as $n \to \infty$. Since $E$ is convex, $\frac{1}{2}(x_n + x_m) \in E$, whence $\|x_n + x_m\| \geqslant 2d$, for all $m, n \geqslant 1$.

Next, using the "parallelogram identity",

$$\|x_n - x_m\|^2 = 2\|x_n\|^2 + 2\|x_m\|^2 - \|x_n + x_m\|^2.$$

Let $\varepsilon > 0$ and choose $N \geqslant 1$ such that whenever $n \geqslant N$,

$$d \leqslant \|x_n\| \leqslant \sqrt{d^2 + \varepsilon^2}.$$

Thus, for $m, n \geqslant N$,

$$\|x_n - x_m\|^2 \leqslant 4d^2 + 4\varepsilon^2 - \|x_n + x_n\|^2 \leqslant 4\varepsilon^2,$$

thus $\|x_n - x_m\| \leqslant 2\varepsilon$ whenever $m, n \geqslant N$. This shows that $(x_n)$ is Cauchy and hence, converges to some $x \in E$. Obviously, $\|x\| = d$.

As for uniqueness, suppose $x, y \in E$ with $\|x\| = \|y\| = d$. Then,

$$0 \leqslant \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2 - \|x + y\|^2 \leqslant 2d^2 + 2d^2 - 4d^2 = 0.$$

Thus, $x = y$, thereby completing the proof. ■

The above theorem fails quite spectacularly for Banach spaces.

**EXAMPLE 8.4.** Let $X = C[0,1]$ the $\mathbb{R}$-vector space of real-valued continuous functions on $[0,1]$ with the supremum norm. Let

$$M = \left\{ f \in X : \int_0^{1/2} f(t)\, dt - \int_{1/2}^1 f(t)\, dt = 1 \right\}.$$

Then, $M$ is a closed convex subset of $X$ but no element of $M$ has minimal norm.

*Proof.* Obviously, $M$ is convex. To see that it is closed, note that the linear functional

$$T : X \to \mathbb{R} \qquad f \mapsto \int_0^{1/2} f(t)\, dt - \int_{1/2}^1 f(t)\, dt$$

is a bounded linear functional, and hence, is continuous. Thus, $M$ is closed too.

Next, for any $f \in M$,

$$1 = \left| \int_0^{1/2} f(t)\, dt - \int_{1/2}^1 f(t)\, dt \right| \leqslant \int_0^1 |f(t)|\, dt \leqslant \|f\|_\infty.$$

We contend that

$$\inf\{\|f\|_\infty : f \in M\} = 1.$$

To see this, fix some $0 < \delta < 1/2$. Define the function

$$f(x) = \begin{cases} 1+\varepsilon & 0 \leqslant x \leqslant \frac{1}{2} - \delta \\ \frac{1+\varepsilon}{\delta}\left(\frac{1}{2} - x\right) & \frac{1}{2} - \delta \leqslant x \leqslant \frac{1}{2} + \delta \\ -(1+\varepsilon) & \frac{1}{2} + \delta \leqslant x \leqslant 1. \end{cases}$$

Then,

$$\int_0^{1/2} f(t)\, dt - \int_{1/2}^1 f(t)\, dt = (1+\varepsilon)(1-2\delta) + \delta(1+\varepsilon) = (1-\delta)(1+\varepsilon).$$

Choosing

$$\varepsilon = \frac{\delta}{1-\delta},$$

we get $Tf = 1$. Note that $\|f\|_\infty = 1+\varepsilon$ and as $\delta \to 0^+$, we get $\|f\|_\infty \to 1^+$. This proves our claim.

Finally, suppose $f \in M$ such that $\|f\|_\infty = 1$. Then,

$$0 = \int_0^{1/2} 1 - f(t)\, dt + \int_{1/2}^1 1 + f(t)\, dt.$$

Since both integrals are non-negative and the functions are continuous, we must have $f(t) = 1$ whenever $0 \leqslant t \leqslant 1/2$ and $f(t) = -1$ whenever $1/2 \leqslant t \leqslant 1$, a contradiction. This completes the proof. ■

**THEOREM 8.5.** Let $M$ be a closed subspace of a Hilbert space $H$, then $H = M \oplus M^\perp$.

*Proof.* Since

$$M^\perp = \bigcap_{x \in M} \ker(\cdot, x),$$

it is a closed subspace of $H$. Obviously, $M \cap M^\perp = \{0\}$. It remains to show that $H = M + M^\perp$. Indeed, let $x \in H$ and let $x_1 \in M$ be the unique element minimizing the distance to $x$. We contend that $x_2 = x - x_1$ is perpendicular to $x_1$.

Indeed, note that for every $y \in M$, we have

$$\|x_2\|^2 \leqslant \|x_2 + y\|^2 \implies \|y\|^2 + 2\Re(x_2, y) \geqslant 0,$$

for all $y \in M$. Suppose $(x_2, y) \neq 0$ for some $y \in M$. We can choose $y$ such that $\Re(x_2, y) = -|(x_2, y)|$. Then, replacing $y$ by $\alpha y$ for some $\alpha > 0$, we have $\alpha^2 \|y\|^2 - 2\alpha|(x, y)| \geqslant 0$ for all $\alpha > 0$. This is obviously false, and hence, $(x_2, y) \neq 0$ for all $y \in M$, thereby completing the proof. ■

The above theorem fails for closed subspaces of Banach spaces.

**EXAMPLE 8.6.** $c_0 \subseteq \ell^\infty$ is not complemented.

*Proof.* We begin with a claim.
**Claim.** Let $T : \ell^\infty \to \ell^\infty$ be a bounded linear operator with $c_0 \subseteq \ker T$. Then there is an infinite subset $A \subseteq \mathbb{N}$ such that $Tx = 0$ whenever $x$ is supported in $A$.
**Proof of Claim:** Suppose not. Then, for every infinite subset $A \subseteq \mathbb{N}$, there is an $x \in \ell^\infty$, supported in $A$ such that $Tx \neq 0$. Choose an uncountable collection $\{A_i : i \in I\}$ of infinite subsets of $\mathbb{N}$ with pairwise finite intersections. According to our assumption, there are $x_i \in \ell^\infty$ supported in $A_i$ with $Tx_i \neq 0$ and $\|x_i\| = 1$.

Since $I$ is uncountable, there is an $n \in \mathbb{N}$ such that

$$I_n = \{i \in I : (Tx_i)(n) \neq 0\}$$

is uncountable (because the union of all the $I_n$'s is $I$). Further, there is a positive integer $k$ such that

$$I_{n,k} = \left\{ i \in I : |(Tx_i)(n)| \geqslant \frac{1}{k} \right\}$$

is uncountable (because the union of all the $I_{n,k}$'s is $I_n$).

Let $J \subseteq I_{n,k}$ be finite and set

$$y = \sum_{j \in J} \operatorname{sgn}\big((Tx_j)(n)\big) \cdot x_j.$$

Then,

$$(Ty)(n) = \sum_{j \in J} \operatorname{sgn}\big((Tx_j)(n)\big) \cdot (Tx_j)(n) \geqslant \sum_{j \in J} \frac{1}{k} = \frac{|J|}{k}.$$

Note that for $i \neq j$, $A_i \cap A_j$ is finite and hence, we can write $y = x + z$, where $x$ has finite support and $\|z\| \leqslant 1$. Thus, $x \in c_0 \subseteq \ker T$ and hence,

$$\frac{|J|}{k} \leqslant \|Ty\| = \|Tx + Tz\| = \|Tz\| \leqslant \|T\| \implies |J| \leqslant k\|T\|,$$

which is absurd, since $I_{n,k}$ is infinite. This proves the claim. □

Coming back, suppose $c_0$ were complemented in $\ell^\infty$. Then, there would be a projection operator $P : \ell^\infty \to c_0 \subseteq \ell^\infty$. Set $T = \mathbf{id} - P$. Since $c_0 \subseteq \ker T$, due to the claim above, there is an infinite subset $A \subseteq \mathbb{N}$, such that $Tx = 0$ whenever $x$ is supported in $A$. Consider $\chi_A \in \ell^\infty$, the characteristic function of the set $A$. But note that

$$P(\chi_A) = (\mathbf{id} - T)(\chi_A) = \chi_A \notin c_0,$$

a contradiction. This completes the proof. ■

**Theorem 8.7 (Riesz Representation Lemma).** Let $H$ be a Hilbert space. The natural map $H \to H^*$ given by $y \mapsto (\cdot, y)$ is an isometric and surjective.

*Proof.* Obviously, the map is injective and linear. To see isometry, note that $(y, y) = \|y\|^2$, whence $\|(\cdot,, y)\| \geqslant \|y\|$ and due to Cauchy-Schwarz,

$$|(x, y)| \leqslant \|x\|\|y\| \implies \|(\cdot, y)\| \leqslant \|y\| \implies \|(\cdot, y)\| = \|y\|.$$

It remains to show surjectivity. Let $\Lambda \neq 0$ be a continuous linear functional on $H$ and $N = \ker \Lambda$. Since $N$ is closed, we can write $H = N \oplus N^\perp$. Choose a nonzero vector $z \in N^\perp$. For any $x \in H$,

$$x - \frac{\Lambda x}{\Lambda z} z \in \ker \Lambda,$$

whence

$$0 = \left(x - \frac{\Lambda x}{\Lambda z} z, z\right) = (x, z) - \frac{\Lambda x}{\Lambda z} \|z\|^2.$$

Thus,

$$\Lambda x = \left(x, \frac{\overline{\Lambda z}}{\|z\|^2} z\right),$$

thereby completing the proof. ■

**THEOREM 8.8.** Let $H$ be a Hilbert space and suppose $f : H \times H \to \mathbb{K}$ is sesquilinear and bounded, that is,

$$M := \sup\{|f(x,y)| : \|x\| = \|y\| = 1\} < \infty,$$

then there exists a unique $S \in \mathscr{B}(H)$ such that

$$f(x,y) = (x, Sy) \quad \forall x, y \in H.$$

Further, $\|S\| = M$.

*Proof.* Fix $y \in H$ and consider the mapping $x \mapsto f(x,y)$. This is a continuous linear functional on $H$ and hence, is given by $x \mapsto (x, Sy)$ for a unique $Sy \in H$. We claim that the association $y \mapsto Sy$ is linear.

Indeed, if $y_1, y_2 \in H$, then

$$f(\cdot, y_1 + y_2) = f(\cdot, y_1) + f(\cdot, y_2) = f(\cdot, Sy_1) + f(\cdot, Sy_2) = f(\cdot, Sy_1 + Sy_2).$$

Due to uniqueness of $S(y_1 + y_2)$, we see that $S(y_1 + y_2) = Sy_1 + Sy_2$. Next, let $\alpha \in \mathbb{K}$ and $y \in H$. Then,

$$(\cdot, S(\alpha y)) = f(\cdot, \alpha y) = \overline{\alpha} f(\cdot, y) = \overline{\alpha}(\cdot, Sy) = (\cdot, \alpha Sy),$$

whence $S(\alpha y) = \alpha Sy$, i.e., $S$ is linear.

Finally, we must show that $\|S\| = M$. Indeed, for $\|x\| = \|y\| = 1$:

$$|f(x,y)| \leqslant |(x, Sy)| \leqslant \|x\| \|Sy\| \leqslant \|S\|,$$

whence $M \leqslant \|S\|$. On the other hand, if $Sy \neq 0$, then

$$\|Sy\| = \left(\frac{Sy}{\|Sy\|}, Sy\right) = f\left(\frac{Sy}{\|Sy\|}, y\right) \leqslant M$$

Taking supremum over $\|y\| = 1$, we have that $\|S\| \leqslant M \leqslant \|S\|$, thereby completing the proof. ■

## §§ Adjoints

**DEFINITION 8.9.** Let $T \in \mathscr{B}(H)$. The map $f : H \times H \to \mathbb{K}$ given by $f(x,y) = (Tx, y)$, is a bounded sesquilinear form on $H$, whence, there is a $T^* \in \mathscr{B}(H)$ such that

$$(Tx, y) = f(x,y) = (x, T^* y) \quad \forall x, y \in H.$$

Next, note that

$$(x, Ty) = \overline{(y, T^* x)} = (T^* x, y) = (x, T^{**} y) \quad \forall x, y \in H.$$

Hence, $T^{**} = T$. On the other hand,

$$\|T^*\| = \sup\{|(Tx, y)| : \|x\| = \|y\| = 1\} \leqslant \|T\|.$$

Consequently, $\|T\| = \|T^{**}\| \leqslant \|T^*\| \leqslant \|T\|$, whence, $\|T^*\| = \|T\|$.

Similarly, the following identities are easy to show for $S, T \in \mathscr{B}(H)$:

$$(S+T)^* = S^* + T^*, \quad (\alpha S)^* = \overline{\alpha} S^*, \quad \text{and} \quad (ST)^* = T^* S^*.$$

Therefore,

$$\|Tx\|^2 = (Tx, Tx) = (x, T^*Tx) \leqslant \|T^*T\| \|x\|^2 \quad \forall x \in H.$$

Hence, $\|T\|^2 \leqslant \|T^*T\| \leqslant \|T^*\| \|T\| = \|T\|^2$, whence $\|T\|^2 = \|T^*T\|$. This makes $\mathscr{B}(H)$ a C*-algebra.

## §§ Compact Self-Adjoint Operators

**Lemma 8.10.** Let $H$ be a Hilbert space and $T \in \mathscr{B}(H)$ a compact self-adjoint operator. Then

$$\|T\| = \sup\{|\langle Tx, x\rangle| : \|x\| = 1\}.$$

*Proof.* Let $B$ denote the quantity on the right hand side. Due to the Cauchy-Schwarz Inequality, $B \leqslant \|T\|$. Let $x \neq 0$ and set $\lambda = \sqrt{\frac{\|Tx\|}{\|x\|}}$.

We have

$$\begin{aligned}
\langle Tx, Tx\rangle &= \frac{1}{4}\left|\langle T(\lambda x + \lambda^{-1}Tx), \lambda x + \lambda^{-1}Tx\rangle - \langle T(\lambda x - \lambda^{-1}Tx), \lambda x - \lambda^{-1}Tx\rangle\right| \\
&\leqslant \frac{1}{4}\left|\langle T(\lambda x + \lambda^{-1}Tx), \lambda x + \lambda^{-1}Tx\rangle\right| + \frac{1}{4}\left|\langle T(\lambda x + \lambda^{-1}Tx), \lambda x + \lambda^{-1}Tx\rangle\right| \\
&\leqslant \frac{B}{4}\left(\|\lambda x + \lambda^{-1}Tx\|^2 + \|\lambda x - \lambda^{-1}Tx\|^2\right) \\
&= \frac{B}{2}\left(\|\lambda x\|^2 + \|\lambda^{-1}Tx\|^2\right) \\
&= B\|x\|\|Tx\|.
\end{aligned}$$

Thus, $\|Tx\| \leqslant B\|x\|$, whence $\|T\| \leqslant B$, thereby completing the proof. ■

**Lemma 8.11.** With the notation of the preceding lemma, either $\|T\|$ or $-\|T\|$ is an eigenvalue of $T$.

*Proof.* Due to the preceding lemma, there is a sequence of unit vectors $(x_n)$ in $H$ such that $|\langle Tx_n, x_n\rangle| \to \|T\|$. Since $T$ is self-adjoint,

$$\overline{\langle Tx, x\rangle} = \langle x, Tx\rangle = \langle Tx, x\rangle,$$

and hence, $\langle Tx, x\rangle \in \mathbb{R}$. Therefore, moving to a subsequence, we may suppose that $\langle Tx_n, x_n\rangle \to \lambda \in \{\pm\|T\|\}$. Further, since $T$ is compact, we may replace $(x_n)$ with a subsequence such that $Tx_n \to \lambda y$ for some $y \in H$.

We contend that $x_n \to y$. First, note that

$$|\langle Tx_n, x_n\rangle| \leqslant \|Tx_n\|\|x_n\| = \|Tx_n\| \leqslant \|T\| = |\lambda|.$$

By our choice of the sequence $(x_n)$, $|\langle Tx_n, x_n\rangle| \to |\lambda|$ and hence, $\|Tx_n\| \to |\lambda|$. Next,

$$\begin{aligned}
\|\lambda x_n - Tx_n\|^2 &= \langle \lambda x_n - Tx_n, \lambda x_n - Tx_n\rangle \\
&= \lambda^2 + \|Tx_n\|^2 - \langle \lambda x_n, Tx_n\rangle - \langle Tx_n, \lambda x_n\rangle \\
&= \lambda^2 + \|Tx_n\|^2 - 2\lambda\langle Tx_n, x_n\rangle
\end{aligned}$$

which goes to 0 as $n \to \infty$. Hence, $\|\lambda x_n - Tx_n\| \to 0$ as $n \to \infty$, consequently, $x_n \to y$, thereby completing the proof. ■

# §9 Banach Algebras

**Definition 9.1.** A *Banach algebra* is a $\mathbb{C}$-algebra $\mathscr{A}$ equipped with a norm $\|\cdot\| : \mathscr{A} \to [0,\infty)$ with respect to which it is a Banach space and

$$\|xy\| \leqslant \|x\|\|y\| \quad \forall x, y \in \mathscr{A}.$$

The Banach algebra is said to be *unital* if it possesses a multiplicative identity.

An *involution* on an algebra $\mathscr{A}$ is a map

$$\mathscr{A} \to \mathscr{A} \quad x \mapsto x^*$$

of order 2 that satisfies

$$(x+y)^* = x^* + y^* \quad (\lambda x)^* = \overline{\lambda} x^* \quad (xy)^* = y^* x^*.$$

An algebra equipped with such an involution is called a *$*$-algebra*. A Banach $*$-algebra that satisfies

$$\|x^* x\| = \|x\|^2 \quad \forall x \in \mathscr{A}$$

is called a *$C^*$-algebra*.

**Remark 9.2.** If $\mathscr{A}$ is a $C^*$-algebra, for $x \neq 0$, we have

$$\|x\|^2 = \|x^* x\| \leqslant \|x^*\|\|x\| \implies \|x\| \leqslant \|x^*\| \leqslant \|x^{**}\| = \|x\|,$$

whence $\|x\| = \|x^*\|$. That is, the involution is an isometry.

**Definition 9.3.** If $\mathscr{A}$ and $\mathscr{B}$ are Banach algebras, a *homomorphism* is a bounded linear map $\phi : \mathscr{A} \to \mathscr{B}$ such that $\phi(xy) = \phi(x)\phi(y)$ for all $x, y \in \mathscr{A}$.

Further, if $\mathscr{A}$ and $\mathscr{B}$ are Banach $*$-algebras, a *$*$-homomorphism* is a homomorphism of Banach algebras $\phi : \mathscr{A} \to \mathscr{B}$ such that $\phi(x^*) = \phi(x)^*$ for all $x \in \mathscr{A}$.

**Theorem 9.4.** Let $\mathscr{A}$ be a unital Banach algebra.

(a) If $|\lambda| > \|x\|$, then $\lambda - x$ is invertible in $\mathscr{A}$.

(b) If $x$ is invertible, and $\|y\| < \|x^{-1}\|^{-1}$, then $x - y$ is invertible with inverse

$$(x-y)^{-1} = \sum_{n \geqslant 0} (x^{-1}y)^n x^{-1}.$$

(c) If $x$ is invertible and $\|y\| < \frac{1}{2}\|x^{-1}\|^{-1}$, then

$$\|(x-y)^{-1} - x^{-1}\| < 2\|x^{-1}\|^2\|y\|.$$

(d) $\mathscr{A}^\times \subseteq \mathscr{A}$ is open and $x \mapsto x^{-1}$ on $\mathscr{A}^\times$ is continuous.

*Proof.* (a) We have

$$(\lambda - x)^{-1} = \lambda^{-1}\left(e - \lambda^{-1}x\right)^{-1} = \sum_{n \geqslant 0} \lambda^{-(n+1)} x^{-n},$$

which converges because things are Cauchy and all the good stuff.

(b) Again, we can write

$$(x-y)^{-1} = \left(x(e - x^{-1}y)\right)^{-1} = (e - x^{-1}y)^{-1}x^{-1} = \sum_{n \geqslant 0} (x^{-1}y)x^{-1}.$$

(c) Using the above expansion, we can write

$$\|(x-y)^{-1}-x^{-1}\| \leqslant \sum_{n\geqslant 0} \|x^{-1}\|^{n+2}\|y\|^{n+1} < 2\|x^{-1}\|^2\|y\|.$$

(d) Due to part (b), $\mathscr{A}^\times$ is open in $\mathscr{A}$ and due to part (c), $x \mapsto x^{-1}$ is continuous.

■

**Definition 9.5.** Let $\mathscr{A}$ be a unital Banach algebra and $x \in \mathscr{A}$. The *spectrum* of $x$ is

$$\sigma(x) = \{\lambda \in \mathbb{C} \colon \lambda e - x \text{ is not invertible}\}.$$

For $\lambda \notin \sigma(x)$, define the *resolvent* of $x$ as

$$R_x(\lambda) = (\lambda e - x)^{-1} : \mathbb{C} \setminus \sigma(x) \to \mathscr{A}.$$

**Proposition 9.6.** For any $x \in \mathscr{A}$, $\sigma(x)$ is a compact subset of $\mathbb{C}$ that is contained in the disk $\{\lambda \in \mathbb{C} \colon |\lambda| \leqslant \|x\|\}$.

*Proof.* Obviously, if $|\lambda| > \|x\|$, then $\lambda e - x$ is invertible. Thus, $\sigma(x)$ is contained in the above disk. Consider the map $\lambda \mapsto \lambda e - x$, which is continuous and hence, the preimage of $\mathscr{A}^\times$ is open in $\mathbb{C}$. As a result, $\sigma(x)$ is closed, thereby completing the proof. ■

**Proposition 9.7.** $R_x$ is an analytic function. And, $R_x(\lambda) \to 0$ as $\lambda \to \infty$.

*Proof.* We have

$$\begin{aligned} R_x(\mu) - R_x(\lambda) &= (\mu e - x)^{-1} - (\lambda e - x)^{-1} \\ &= R_x(\mu)\big((\lambda e - x) - (\mu e - x)\big) R_x(\lambda). \end{aligned}$$

Hence,

$$\frac{R_x(\mu) - R_x(\lambda)}{\mu - \lambda} = -R_x(\mu) R_x(\lambda).$$

In the limit $\mu \to \lambda$, we get

$$R_x'(\lambda) = -R_x(\lambda)^2.$$

As for the second part, simply note that for $|\lambda| > \|x\|$,

$$\|R_x(\lambda)\| = \left\| \sum_{n\geqslant 0} \lambda^{-(n+1)} x^n \right\| \leqslant |\lambda|^{-1} \sum_{n\geqslant 0} |\lambda|^{-n} \|x\|^n = \frac{1}{|\lambda| - \|x\|},$$

which goes to 0 as $\lambda \to \infty$, thereby completing the proof. ■

**Theorem 9.8 (Gelfand-Mazur).** Let $\mathscr{A}$ be a unital Banach algebra $\sigma(x) \neq \emptyset$ for every $x \in \mathscr{A}$.

*Proof.* Suppose $\sigma(x) = \emptyset$ for some $x \in \mathscr{A}$. Then, $R_x : \mathbb{C} \to \mathscr{A}$ is an analytic function. For any $\Lambda \in \mathscr{A}^*$, $\Lambda \circ R_x$ is an entire function and is bounded, since

$$\lim_{\lambda\to\infty} \Lambda(R_x(\lambda)) = \Lambda\left(\lim_{\lambda\to\infty} R_x(\lambda)\right) = 0.$$

Due to Liouville's Theorem, $\Lambda \circ R_x$ must be constant on $\mathbb{C}$ and equal to 0. Since this is true for every $\Lambda \in \mathscr{A}^*$, we see that $R(\lambda) = 0$ for every $\lambda \in \mathbb{C}$, which is absurd. This completes the proof. ■

**COROLLARY.** If $\mathscr{A}$ is a unital Banach algebra in which every nonzero element is invertible, then $\mathscr{A} = \mathbb{C}e$.

*Proof.* Suppose $x \in \mathscr{A} \setminus \mathbb{C}e$, then $\lambda e - x \neq 0$ for every $\lambda \in \mathbb{C}$, whence, $\lambda e - x$ is invertible for every $\lambda \in \mathbb{C}$, a contradiction. ■

**DEFINITION 9.9.** Let $\mathscr{A}$ be a unital Banach algebra. For $x \in \mathscr{A}$, the *spectral radius* of $x$ is defined to be

$$\rho(x) := \sup\{|\lambda| : \lambda \in \sigma(x)\}.$$

We have the obvious inequality $\rho(x) \leqslant \|x\|$.

**THEOREM 9.10 (SPECTRAL RADIUS FORMULA).** Let $\mathscr{A}$ be a unital Banach algebra and $x \in \mathscr{A}$. Then,

$$\rho(x) = \lim_{n\to\infty} \|x^n\|^{1/n}.$$

*Proof.* If $\lambda \in \sigma(x)$, then

$$\lambda^n e - x^n = (\lambda e - x)\left(\lambda^{n-1}e + \cdots + x^{n-1}\right).$$

Consequently, $\lambda^n e - x^n$ cannot be invertible. Hence, $|\lambda|^n \leqslant \|x^n\|$. In particular, this gives

$$\rho(x) \leqslant \liminf_{n\to\infty} \|x^n\|^{1/n}.$$

Next, for $|\lambda| > \|x\|$, we have a Laurent series about infinity:

$$\Lambda \circ R_x(\lambda) = \sum_{n\geqslant 0} \lambda^{-(n+1)} \Lambda(x^n).$$

Note that $\Lambda \circ R_x$ is analytic on $|\lambda| > \rho(x)$ and hence, the above Laurent series must be valid there too.

Hence, for any $|\lambda| > \rho(x)$, there is a constant $C_\Lambda > 0$ such that

$$|\Lambda(\lambda^{-n}x^n)| = |\lambda^{-n}\Lambda(x^n)| \leqslant C_\Lambda \quad \forall n \in \mathbb{N}.$$

This holds for all $\Lambda \in \mathscr{A}^*$. Thus, the sequence $(\lambda^{-n}x^n)$ is bounded, that is, there is a $C > 0$ such that $\|x^n\| \leqslant C|\lambda|^n$. Hence,

$$\limsup_{n\to\infty} \|x^n\|^{1/n} \leqslant \limsup_{n\to\infty} C^{1/n}|\lambda| = |\lambda|.$$

Taking infimum over $\lambda$, we get

$$\limsup_{n\to\infty} \|x^n\|^{1/n} \leqslant \rho(x) \leqslant \liminf_{n\to\infty} \|x^n\|^{1/n},$$

thereby completing the proof. ■

**DEFINITION 9.11.** Let $\mathscr{A}$ be a unital Banach algebra. A *multiplicative functional* on $\mathscr{A}$ is a *nonzero* homomorhpism $h : \mathscr{A} \to \mathbb{C}$. The set of all multiplicative functionals on $\mathscr{A}$ is called the *spectrum* of $\mathscr{A}$ and is denoted by $\sigma(\mathscr{A})$.

**PROPOSITION 9.12.** Let $\mathscr{A}$ be a unital Banach algebra and suppose $h \in \sigma(\mathscr{A})$.

(a) $h(e) = 1$.

(b) If $x \in \mathscr{A}^\times$, then $h(x) \neq 0$.

(c) $|h(x)| \leqslant \rho(x) \leqslant \|x\|$ for all $x \in \mathscr{A}$. That is, $\|h\| \leqslant 1$.

*Proof.* (a) Since $h \neq 0$, there is an $x \in \mathscr{A}$ such that $h(x) \neq 0$. Then,

$$h(x) = h(xe) = h(x)h(e) \implies h(e) = 1.$$

(b) Obviously,

$$1 = h(e) = h(x^{-1}x) = h(x^{-1})h(x) \implies h(x) \neq 0.$$

(c) Suppose $|\lambda| > \rho(x)$. Then, $\lambda e - x \in \mathscr{A}^\times$, consequently,

$$0 \neq h(\lambda e - x) = \lambda - h(x) \implies h(x) \neq \lambda.$$

Since this holds for all $|\lambda| > \rho(x)$, we have $|h(x)| \leqslant \rho(x) \leqslant \|x\|$. ■

As a consequence, $\sigma(\mathscr{A})$ is contained in the closed unit ball of $\mathscr{A}^*$. Equip the latter with the weak*-topology. Using nets, it is easy to see that $\sigma(\mathscr{A})$ is closed in $\mathscr{A}^*$. Due to Theorem 4.12, the closed unit ball of $\mathscr{A}^*$ is weak*-compact and hence, so is $\sigma(\mathscr{A})$ with the subspace topology from the weak*-topology on $\mathscr{A}^*$. Thus, $\sigma(\mathscr{A})$ is a *compact Hausdorff space*.

**PROPOSITION 9.13.** Let $\mathscr{A}$ be a commutative unital Banach algebra and $\mathscr{J} \subsetneq \mathscr{A}$ be a proper ideal.

(a) $\mathscr{J} \subseteq \mathscr{A} \setminus \mathscr{A}^\times$

(b) $\overline{\mathscr{J}}$ is a proper ideal.

(c) $\mathscr{J}$ is contained in a maximal ideal.

(d) Every maximal ideal is closed.

*Proof.* The first assertion is obvious. As for the second, note that $\mathscr{A} \setminus \mathscr{A}^\times$ is closed and hence, $\overline{\mathscr{J}} \subseteq \mathscr{A} \setminus \mathscr{A}^\times$. Consequently, $\overline{\mathscr{J}} \neq \mathscr{A}$. To see that it is an ideal, suppose $x \in \overline{\mathscr{J}}$ and $a \in \mathscr{A}$. Then, there is a sequence $(x_n)$ converging to $x$. Consequently, $(ax_n)$ converges to $ax$. But each $ax_n \in \mathscr{J}$ and hence, $ax \in \overline{\mathscr{J}}$. This proves (b).

The third assertion is a standard application of Zorn's lemma. As for (d), if $\mathscr{M}$ is a maximal ideal, then $\mathscr{M} \subseteq \overline{\mathscr{M}} \subsetneq \mathscr{A}$ due to (b). The maximality of $\mathscr{M}$ forces $\mathscr{M} = \overline{\mathscr{M}}$, thereby completing the proof. ■

**THEOREM 9.14.** Let $\mathscr{A}$ be a commutative unital Banach algebra. Then, the map $h \mapsto \ker h$ is a bijective correspondence between $\sigma(\mathscr{A})$ and the set of all maximal ideals in $\mathscr{A}$.

*Proof.* The map is obviously an injection. We establish surjectivity. Let $\mathscr{M}$ be a maximal ideal in $\mathscr{A}$ and consider the quotient algebra $\mathscr{A}/\mathscr{M}$ equipped with the norm:

$$\|x + \mathscr{M}\| = \inf\{\|x + y\| : y \in \mathscr{M}\}.$$

This is again a commutative unital Banach algebra in which every non-zero element is invertible (standard fact from ring theory). Due to Gelfand-Mazur, $\mathscr{A}/\mathscr{M} \cong \mathbb{C}(e + \mathscr{M})$. The composition

$$\mathscr{A} \longrightarrow \mathscr{A}/\mathscr{M} \cong \mathbb{C}(e + \mathscr{M}) \cong \mathbb{C}$$

is the required linear functional, thereby proving surjectivity. ■

**DEFINITION 9.15.** Let $\mathscr{A}$ be a commutative unital Banach algebra. For each $x \in \mathscr{A}$, there is a continuous function $\widehat{x} : \sigma(\mathscr{A}) \to \mathbb{C}$ given by $h \mapsto h(x)$. This gives a map

$$\Gamma_{\mathscr{A}} : \mathscr{A} \to C(\sigma(A)) \qquad x \mapsto \widehat{x},$$

known as the *Gelfand transform* on $\mathscr{A}$.

**PROPOSITION 9.16.** Let $\mathscr{A}$ be a commutative unital Banach algebra and $x \in \mathscr{A}$.

(a) The $\Gamma : \mathscr{A} \to C(\sigma(\mathscr{A}))$ is a homomorphism, and $\widehat{e}$ is the constant function 1.

(b) $x$ is invertible if and only if $\widehat{x}$ never vanishes.

(c) The range of $\widehat{x} : \sigma(\mathscr{A}) \to \mathbb{C}$ is precisely $\sigma(x)$.

(d) $\|\widehat{x}\|_{\sup} = \rho(x) \leqslant \|x\|$.

*Proof.* (a) Obvious.

(b) If $x$ is invertible, then due to (a), so is $\widehat{x}$, whence it never vanishes. On the other hand, if $x$ is not invertible, then it is contained in some maximal ideal $\mathfrak{M}$, whence, there is an $h \in \sigma(\mathscr{A})$ that vanishes on $x$. Thus, $\widehat{x}(h) = 0$, that is, $\widehat{x}$ vanishes somewhere.

(c) Next, suppose $\lambda = \widehat{x}(h) = h(x)$. Then, $h(\lambda e - x) = 0$, hence, $\lambda e - x$ is not invertible, i.e. $\lambda \in \sigma(x)$. Similarly, if $\lambda \in \sigma(x)$, then $\lambda e - x$ is not invertible and hence, $\widehat{x}$ vanishes somewhere, consequently, $h(\lambda e - x) = 0$ for some $h \in \sigma(\mathscr{A})$. This shows that $\lambda$ is in the range of $\widehat{x}$.

(d) Follows from (c). ■

**DEFINITION 9.17.** Let $\mathscr{A}$ be a commutative unital Banach $*$-algebra. If $\Gamma : \mathscr{A} \to C(\sigma(\mathscr{A}))$ is a $*$-homomorphism, then $\mathscr{A}$ is said to be *symmetric*.

**REMARK 9.18.** Note that $\mathscr{A}$ being symmetric is the same as saying

$$\widehat{x^*} = \overline{\widehat{x}} \quad \forall x \in \mathscr{A}.$$

**PROPOSITION 9.19.** Let $\mathscr{A}$ be a commutative Banach $*$-algebra.

(a) $\mathscr{A}$ is symmetric if and only if $\widehat{x}$ is real-valued whenever $x = x^*$.

(b) Every C*-algebra is symmetric.

(c) If $\mathscr{A}$ is symmetric, $\Gamma(\mathscr{A})$ is dense in $C(\sigma(\mathscr{A}))$.

*Proof.* (a) If $\mathscr{A}$ is symmetric and $x^* = x$, then $\widehat{x} = \widehat{x^*} = \overline{\widehat{x}}$, whence $\widehat{x}$ is real-valued. Next, we prove the converse. For any $x \in \mathscr{A}$, write

$$x = \underbrace{\frac{x + x^*}{2}}_{y} + \underbrace{\frac{x - x^*}{2}}_{z}.$$

Note that $y^* = y$ and $z + z^* = 0$. Our hypothesis implies $\widehat{y}$ is real-valued and $\widehat{z} + \overline{\widehat{z}} = 0$. Thus,

$$\widehat{x^*} = \widehat{y^*} + \widehat{z^*} = \widehat{y} - \widehat{z} = \widehat{y} + \overline{\widehat{z}} = \overline{\widehat{x}}.$$

(b) Let $x \in \mathscr{A}$ be such that $x^* = x$. Suppose $h(x) = \alpha + i\beta$. We shall show that $\beta = 0$. Indeed, for $t \in \mathbb{R}$, let $z = x + ite$. Then,

$$z^* z = (x - ite)(x + ite) = x^2 + t^2 e.$$

And hence,

$$|\alpha + (\beta + t)i|^2 = |h(z)|^2 \leqslant \|z\|^2 = \|z^* z\| = \|x^2 + t^2 e\| \leqslant \|x\|^2 + t^2.$$

That is,

$$\alpha^2 + 2\beta t + \beta^2 \leqslant \|x\|^2 \quad \forall t \in \mathbb{R}.$$

Thus, $\beta = 0$ and due to (a), $\mathscr{A}$ is symmetric.

(c) Note that $\Gamma(\mathscr{A})$ contains all the constant functions and thus, the family $\Gamma(\mathscr{A})$ does not vanish at any point. Next, by definition, $\Gamma(\mathscr{A})$ separates points. Further, since $\Gamma$ is a $*$-homomorophism, $\Gamma(\mathscr{A})$ is closed under taking conjugates. Thus, $\Gamma(\mathscr{A})$ is dense in $C(\sigma(\mathscr{A}))$ due to the Stone-Weierstrass Theorem. ■

**PROPOSITION 9.20.** Let $\mathscr{A}$ be a commutative unital Banach algebra.

(a) If $x \in \mathscr{A}$, then $\|\widehat{x}\|_{\sup} = \|x\|$ if and only if $\|x^{2^k}\| = \|x\|^{2^k}$ for all $k \geqslant 1$.

(b) $\Gamma : \mathscr{A} \to C(\sigma(\mathscr{A}))$ is an isometry if and only if $\|x^2\| = \|x\|^2$ for all $x \in \mathscr{A}$.

*Proof.* (a) This follows immediately from the spectral radius formula.

$$\|\widehat{x}\|_{\sup} = \rho(x) = \lim_{k\to\infty} \|x^{2^k}\|^{1/2^k} = \lim_{k\to\infty} \|x\|^{2^k \cdot 2^{-k}} = \|x\|.$$

(b) We have

$$\|x^{2^k}\| = \|x^{2^{k-1}}\|^2 = \cdots = \|x\|^{2^k}.$$ ■

**THEOREM 9.21 (GELFAND-NAIMARK).** If $\mathscr{A}$ is a commutative unital C*-algebra, then $\Gamma : \mathscr{A} \to C(\sigma(\mathscr{A}))$ is an isometric $*$-isomorphism.

*Proof.* That $\Gamma$ is a $*$-homomorphism has already been established. We first show that $\Gamma$ is an isometry. Let $x \in \mathscr{A}$ and set $y = x^* x$. Then, $y^* = y$, so

$$\|y^{2^k}\| = \left\| \left(y^{2^{k-1}}\right)^* y^{2^{k-1}} \right\| = \|y^{2^{k-1}}\|^2 = \cdots = \|y\|^{2^k}.$$

Due to part (a) of the preceding result, $\|\widehat{y}\|_{\sup} = \|y\|$. But $\widehat{y} = \overline{\widehat{x}}\widehat{x} = |\widehat{x}|^2$. Hence,

$$\|\widehat{x}\|_{\sup}^2 = \|\widehat{y}\|_{\sup} = \|y\| = \|x\|^2 \implies \|\widehat{x}\|_{\sup} = \|x\|,$$

whence, due to part (b) of the preceding result, $\Gamma$ is an isometry. Thus, its image is closed in $C(\sigma(\mathscr{A}))$. But we already argued that $\Gamma(\mathscr{A})$ is dense in $C(\sigma(\mathscr{A}))$ and hence, $\Gamma$ must be surjective. This completes the proof. ■

# §10 DISTRIBUTIONS

## §§ The topology on $\mathscr{D}_K$

Let $\Omega \subseteq \mathbb{R}^n$ be an open set. We begin by topologizing $C^\infty(\Omega)$. Fix an exhaustion $\{K_i\}$ of $\Omega$ by compact sets. That is,

- $\Omega = \bigcup_{i=1}^{\infty} K_i$, and
- $K_i \subseteq K_{i+1}^\circ$ for all $i \geqslant 1$.

Define the seminorms $p_N : C^\infty(\Omega) \to \mathbb{R}$ given by

$$p_N(\phi) = \sup\left\{|\partial^\alpha \phi(x)| : x \in K_N,\ |\alpha| \leqslant N\right\}.$$

That this is a separating family of seminorms is obvious, and since this is a countable family, the induced locally convex vector topology on $C^\infty(\Omega)$ is metrizable.

It is easy to see that the "evaluation functionals" on $C^\infty(\Omega)$ equipped with this topology are continuous, therefore,

$$\mathscr{D}_K := \bigcap_{x\in\Omega\setminus K} \ker \mathrm{ev}_x$$

is closed in $C^\infty(\Omega)$. It is easy to see that a (countable) local base at 0 is given by the sets

$$V_N = \left\{ f \in C^\infty(\Omega) \colon p_N(f) < \frac{1}{N} \right\}$$

for $N \geqslant 1$. Further, in this topology $C^\infty(\Omega)$ is a Fréchet space[1] and since $\mathscr{D}_K$ is closed, it to is a Fréchet space. It can also be showed that $C^\infty(\Omega)$ has the Heine-Borel property and hence, the same conclusion holds for $\mathscr{D}_K$.

## §§ Distributions

Let $\Omega \subseteq \mathbb{R}^n$ be an open set. Define

$$\mathscr{D}(\Omega) = \bigcup_{K \Subset \Omega} \mathscr{D}_K.$$

Introduce the seminorms $\|\cdot\|_N : \mathscr{D}(\Omega) \to \mathbb{R}$ given by

$$\|\phi\|_N = \max\left\{ |\partial^\alpha \phi(x)| \colon x \in \Omega,\ |\alpha| \leqslant N \right\},$$

for $\phi \in \mathscr{D}(\Omega)$ and $N \geqslant 0$. The restrictions of these seminorms to $\mathscr{D}_K$ are still seminorms. We claim that they induce the same topology as the canoical topology of $\mathscr{D}_K$ discussed in the preceding (sub)section. First, there is a positive integer $N_0$ such that $K \subseteq K_N$ for all $N \geqslant N_0$. For these $N$, $\|\phi\|_N = p_N(\phi)$ if $\phi \in \mathscr{D}_K$. Further, since $\|\phi\|_N \leqslant \|\phi\|_{N+1}$, the topologies induced by either sequence of seminorms are unchanged if we let $N$ start at $N_0$ instead of 1. Thus, the two topologies coincide and a local base is given by sets of the form

$$V_N = \left\{ \phi \in \mathscr{D}_K \colon \|\phi\|_N < \frac{1}{N} \right\}.$$

In particular, $\mathscr{D}_K$ is still a Fréchet space having the Heine-Borel property.

**Definition 10.1.** Let $\Omega \subseteq \mathbb{R}^n$ be a nonempty open set.

(a) For every compact $K \Subset \Omega$, let $\tau_K$ denote the standard Fréchet space topology of $\mathscr{D}_K$.

(b) Let $\beta$ denote the collection of all convex balanced sets $W \subseteq \mathscr{D}(\Omega)$ such that $\mathscr{D}_K \cap W \in \tau_K$ for every $K \Subset \Omega$.

(c) $\tau$ is the collection of all unions of sets of the form $\phi + W$, with $\phi \in \mathscr{D}(\Omega)$ and $W \in \beta$.

**Theorem 10.2.** (a) $\tau$ is a topology on $\mathscr{D}(\Omega)$, and $\beta$ is a local base for $\tau$.

(b) $\tau$ makes $\mathscr{D}(\Omega)$ into a locally convex topological vector space.

*Proof.* (a) Let $V_1, V_2 \in \tau$. We shall show that for all $\phi \in V_1 \cap V_2$, there is some $W \in \beta$ such that $\phi + W \subseteq V_1 \cap V_2$. Since each $V_i$ is open, there is some $W_i \in \beta$ such that $\phi \in \phi_i + W_i \subseteq$. Let $K \Subset \Omega$ such that $\phi, \phi_1, \phi_2 \in \mathscr{D}_K$. Since each $\mathscr{D}_K \cap W_i$ is open in $\mathscr{D}_K$, $W_i$ is convex and balanced, and $\phi - \phi_i \in \mathscr{D}_K \cap W_i$. Since the Minkowski functional on $\mathscr{D}_K$ corresponding to $\mathscr{D}_K \cap W_i$ is continuous, there is a $0 < \delta_i < 1$ such that $\phi - \phi_i \in (1-\delta_i)W_i$. Hence,

$$\phi - \phi_i + \delta_i W_i \subseteq (1-\delta_i) \subseteq W_i \implies \phi + \delta_i W_i \subseteq \phi_i + W_i \subseteq V_i,$$

whence $\phi + (\delta_1 W_1) \cap (\delta_2 W_2) \subseteq V_1 \cap V_2$. Since $\delta_1 W_1 \cap \delta_2 W_2 \in \beta$, the conclusion follows.

[1] I might add in the details to this some day.

(b) Since $\beta$ consists of convex sets, it suffices to show that $\tau$ makes $\mathscr{D}(\Omega)$ a topological vector space. First, we must show that the space is $T_1$. Let $\phi_1 \neq \phi_2 \in \mathscr{D}(\Omega)$, and set

$$W = \{\phi \in \mathscr{D}(\Omega) \colon \|\phi\|_0 < \|\phi_1 - \phi_2\|_0\},$$

where $\|\cdot\|_0$ is precisely the sup-norm on $\Omega$. By definition, it is easy to see that $W \in \beta$ and $\phi_1 \notin \phi_2 + W$, consequently, $\{\phi_1\}$ is closed.

To see that addition is continuous, let $(\phi_1, \phi_2) \mapsto \phi_1 + \phi_2$ and $V$ an open set containing $\phi_1 + \phi_2$. Since $\beta$ forms a local base for the topology, we can find some $W \in \beta$ such that $(\phi_1 + \phi_2) + W \subseteq V$, and

$$\left(\phi_1 + \frac{1}{2}W\right) + \left(\phi_2 + \frac{1}{2}W\right) \subseteq (\phi_1 + \phi_2) + W \subseteq V.$$

Thus, addition is continuous.

Finally, we must show that scalar multiplication is continuous. Let $\alpha_0 \in \mathbb{K}$ and $\phi_0 \in \mathscr{D}(\Omega)$. Then,

$$\alpha\phi - \alpha_0\phi_0 = \alpha(\phi - \phi_0) + (\alpha - \alpha_0)\phi_0.$$

Let $V$ be an open set containing $\alpha_0\phi_0$, and choose a $W \in \beta$ such that $\alpha_0\phi_0 + W \subseteq V$. There is a $\delta > 0$ such that $\delta\phi_0 \in \frac{1}{2}W$. Next, choose $c > 0$ such that $2c(|\alpha_0| + \delta) = 1$. For $|\alpha - \alpha_0| < \delta$ and $\phi - \phi_0 \in cW$, we have

$$\alpha\phi - \alpha_0\phi_0 \in |\alpha| cW + \frac{1}{2}W \subseteq c(|\alpha_0| + \delta)W + \frac{1}{2}W \subseteq W,$$

as desired. This completes the proof. ■

**THEOREM 10.3.** (a) A convex balanced subset $V$ of $\mathscr{D}(\Omega)$ is open if and only if $V \in \beta$.

(b) The topology $\tau_K$ of any $\mathscr{D}_K \subseteq \mathscr{D}(\Omega)$ coincides with the subspace topology that $\mathscr{D}_K$ inherits from $\mathscr{D}(\Omega)$.

(c) If $E$ is a bounded subset of $\mathscr{D}(\Omega)$, then $E \subseteq \mathscr{D}_K$ for some $K \subseteq \Omega$ and there are real numbers $0 < M_N < \infty$ such that every $\phi \in E$ satisfies the inequalities $\|\phi\|_N \leqslant M_N$ for $N \geqslant 0$.

(d) $\mathscr{D}(\Omega)$ has the Heine-Borel property, that is, closed and bounded sets are compact.

(e) If $\{\phi_i\}$ is a Cauchy sequence in $\mathscr{D}(\Omega)$, then $\{\phi_i\} \subseteq \mathscr{D}_K$ for some compact $K \subseteq \Omega$, and

$$\lim_{(i,j)\to\infty} \|\phi_i - \phi_j\|_N = 0$$

for all $N \geqslant 0$.

(f) If $\phi_i \to 0$ in the topology of $\mathscr{D}(\Omega)$, then there is a compact set $K \subseteq \Omega$ which contains the support of every $\phi_i$ and $\partial^\alpha \phi_i \to 0$ uniformly as $i \to \infty$, for every multi-index $\alpha$.

(g) $\mathscr{D}(\Omega)$ is a Fréchet space.

*Proof.* Let $V \in \tau$ and $\phi \in \mathscr{D}_K \cap V$. Since $\beta$ form a local base, there is a $W \in \beta$ such that $\phi + W \subseteq V$. Hence,

$$\phi + (\mathscr{D}_K \cap W) \subseteq \mathscr{D}_K \cap V.$$

Since $\mathscr{D}_K \cap W$ is open in $\mathscr{D}_K$, we have that $\mathscr{D}_K \cap V \in \tau_K$.

(a) Now, let $V$ be a convex balanced subset of $\mathscr{D}(\Omega)$. If $V$ is open, then due to our observation above, $V \in \beta$. The converse direction is trivial since $\beta \subseteq \tau$.

(b) The above remark shows that the induced topology on $\mathscr{D}_K$ is coarser than $\tau_K$. Conversely, suppose $E \in \tau_K$. We have to show that $E = \mathscr{D}_K \cap V$ for some $V | in \tau$. By definition, for every $\phi \in E$, there is a positive integer $N$ and $\delta > 0$ such that

$$\{\psi \in \mathscr{D}_K : \|\psi - \phi\|_N < \delta\} \subseteq E.$$

Set $W_\phi = \{\psi \in \mathscr{D}(\Omega) : \|\psi\|_N < \delta\} \in \beta$, so that

$$\mathscr{D}_K \cap (\phi + W_\phi) = \phi + \mathscr{D}_K \cap W_\phi \subseteq E.$$

Since $W_\phi \in \beta$ for every $\phi \in E$, we see that $V := \bigcup_{\phi \in E} (\phi + W_\phi)$ is an element of $\tau$ and $V \cap E = E$, as desired.

(c) Suppose $E$ does not lies in any $\mathscr{D}_K$. Using an exhaustion of $\Omega$, we can find a sequence of functions $\phi_m \in E$ and distinct points $x_m \in \Omega$ with no limit point in $\Omega$ such that $\phi_m(x_m) \neq 0$. Let $W$ be the set of all $\phi \in \mathscr{D}(\Omega)$ which satisfy

$$|\phi(x_m)| < \frac{1}{m}|\phi_m(x_m)| \qquad \forall\ m \geqslant 1.$$

Note that

$$W \cap \mathscr{D}_K = \bigcap_{x_m \in W \cap \mathscr{D}_K} \left\{\phi \in \mathscr{D}_K : |\phi(x_m)| < \frac{1}{m}|\phi_m(x_m)|\right\},$$

which is a finite intersection since only finitely many of the $x_m$'s can be contained in $K$ (as they do not admit a limit point in $\Omega$). Thus, $W \cap \mathscr{D}_K$ is open, owing to the continuity of the "evaluation functionals" on $\mathscr{D}_K$; hence $W \in \beta$. Since $\phi_m \notin mW$, no multiple of $W$ contains $E$, which shows that $E$ is not bounded. Hence, every bounded $E$ lies in some $\mathscr{D}_K$. Being a bounded subset of $\mathscr{D}_K$, every seminorm on $\mathscr{D}_K$ is bounded on $E$, whence the last assertion of (c) follows.

(d) This follows immediately from the above parts, since every bounded set is contained in some $\mathscr{D}_K$, whose subspace topology is same as the canonical topology, in which it has the Heine-Borel property.

(e) Every Cauchy sequence is bounded and hence, is contained in some $\mathscr{D}_K$, which has its canonical topology induced by the seminorms $\|\cdot\|_N$, whence the conclusion follows.

(f) This follows immediately from (e).

(g) Finally, we have shown that any Cauchy sequence in $\mathscr{D}(\Omega)$ lies in $\mathscr{D}_K$, which is Fréchet, whence it must converge. This completes the proof. ■

**THEOREM 10.4.** Let $\Lambda$ be a linear map from $\mathscr{D}(\Omega)$ to a locally convex space $Y$. Then the following are equivalent:

(a) $\Lambda$ is continuous.

(b) $\Lambda$ is bounded.

(c) If $\phi_i \to 0$ in $\mathscr{D}(\Omega)$, then $\Lambda\phi_i \to 0$ in $Y$.

(d) The restriction of $\Lambda$ to every $\mathscr{D}_K \subseteq \mathscr{D}(\Omega)$ are continuous.

*Proof.* $(a) \implies (b)$ is well known. Next, if $\phi_i \to 0$ in $\mathscr{D}(\Omega)$, then it is contained in some $\mathscr{D}_K$ for a compact $K \Subset \Omega$. Since the restriction of $\Lambda$ to $\mathscr{D}_K$ is continuous and it is a metrizable topological vector space, $\Lambda\phi_i \to 0$ in $Y$, thereby proving $(b) \implies (c)$.

To see $(c) \implies (d)$, it suffices to show that the restriction of $\Lambda$ to each $\mathscr{D}_K$ is sequentially continuous. If $\phi_i \to 0$ in $\mathscr{D}_K$ and since the topology of $\mathscr{D}_K$ is the subspace topology, we see that $\phi_i \to 0$ in $\mathscr{D}(\Omega)$ and according to our assumption, $\Lambda\phi_i \to 0$ in $Y$, which proves sequential continuity.

Finally, let $U$ be a convex balanced neighborhood of 0 in $Y$. It suffices to show that $V = \Lambda^{-1}U$ is open. Note that $V$ is a convex balanced subset of $\mathscr{D}(\Omega)$ containing 0. Due to Theorem 10.3 (a), $V$ is open in $\mathscr{D}(\Omega)$ if and only if $V \cap \mathscr{D}_K$ is open in $\mathscr{D}_K$ for every compact $K \Subset \Omega$. But this is precisely the content of (d), thereby completing the proof. ■

**Definition 10.5.** A linear functional on $\mathscr{D}(\Omega)$ continuous with respect to the topology $\tau$ is called a *distribution*.

**Theorem 10.6.** If $\Lambda$ is a linear functional on $\mathscr{D}(\Omega)$, the following are equivalent:

(a) $\Lambda \in \mathscr{D}'(\Omega)$.

(b) To every compact $K \Subset \Omega$, corresponds a nonnegative integer $N$ and a constant $C < \infty$ such that

$$|\Lambda\phi| \leqslant C\|\phi\|_N \qquad \forall\ \phi \in \mathscr{D}_K.$$

*Proof.* If $\Lambda \in \mathscr{D}'(\Omega)$, then the restriction of $\Lambda$ to every $\mathscr{D}_K$ is continuous and so is bounded on some neighborhood of the origin, containing an open neighborhood of the form

$$\{\phi \in \mathscr{D}_K : \|\phi\|_N < \frac{1}{N}\},$$

whence (b) follows.

Conversely, suppose (b) holds. Then, as argued above, the restriction of $\Lambda$ to every $\mathscr{D}_K$ is continuous, and due to the preceding theorem, $\Lambda$ is continuous. ■

**Example 10.7.** There are some canonical examples of distributions. If $f \in L^1_{loc}(\Omega)$, then the map $\Lambda_f : \mathscr{D}(\Omega) \to \mathbb{R}$ given by

$$\Lambda_f(\phi) = \int_\Omega f(x)\phi(x)\, dx \qquad \forall\ \phi \in \mathscr{D}(\Omega).$$

If $K = \operatorname{Supp}\phi$, then

$$|\Lambda_f(\phi)| \leqslant \|f\|_{L^1(K)}\|\phi\|_0,$$

and hence $\Lambda_f$ is a distribution of order 0.

**Example 10.8 (Dirac Distribution).** The map $\Lambda : \mathscr{D}(\Omega) \to \mathbb{R}$ defined by $\phi \mapsto \phi(0)$ is a distribution of order 0, since

$$|\Lambda(\phi)| \leqslant \|\phi\|_0.$$

This is known as the *Dirac distribution*.

**Definition 10.9.** Let $\Lambda \in \mathscr{D}'(\Omega)$ be a distribution and $\alpha$ a multi-index. Define $\partial^\alpha\Lambda \in \mathscr{D}'(\Omega)$ by

$$\left(\partial^\alpha\Lambda\right)(\phi) = (-1)^{|\alpha|}\Lambda\left(\partial^\alpha\phi\right).$$

We show that this is indeed a distribution. Let $K \Subset \Omega$ be a compact set. Then there is an integer $N$ and $C > 0$ such that

$$|\Lambda(\phi)| \leqslant C\|\phi\|_N \qquad \forall\ \phi \in \mathscr{D}_K.$$

Then,

$$\left|(\partial^\alpha \Lambda)(\phi)\right| \leqslant C\|\partial^\alpha \phi\|_N \leqslant C\|\phi\|_{N+|\alpha|} \qquad \forall\ \phi \in \mathscr{D}_K,$$

as desired.

We further note that the formula

$$\partial^\alpha \partial^\beta \Lambda = \partial^{\alpha+\beta}\Lambda = \partial^\beta \partial^\alpha \Lambda$$

holds for every distribution $\Lambda \in \mathscr{D}'(\Omega)$. This is quite straightforward, since

$$\left(\partial^\alpha \partial^\beta \Lambda\right)(\phi) = (-1)^{|\alpha|}\partial^\beta \Lambda\left(\partial^\alpha \phi\right) = (-1)^{|\alpha|+|\beta|}\Lambda\left(\partial^{\alpha+\beta}\phi\right) = \left(\partial^{\alpha+\beta}\Lambda\right)(\phi),$$

as desired.

**Definition 10.10.** Let $\Lambda \in \mathscr{D}'(\Omega)$ be a distribution and $f \in C^\infty(\Omega)$. Define $f\Lambda \in \mathscr{D}'(\Omega)$ as

$$(f\Lambda)(\phi) = \Lambda\left(f\phi\right) \qquad \forall\ \phi \in \mathscr{D}_K.$$

We contend that this is indeed a distribution. For any multi-index $\alpha$, we have

$$\partial^\alpha\left(f\phi\right) = \sum_{\substack{\beta+\gamma=\alpha \\ \beta,\gamma \geqslant 0}} \frac{\alpha!}{\beta!\gamma!}(\partial^\beta f)(\partial^\gamma \phi).$$

If $K \Subset \Omega$ is a compact set, then there is a positive integer $N$ and a constant $C > 0$ such that $|\Lambda(f)| \leqslant C\|\phi\|_N$ for all $\phi \in \mathscr{D}_K$. Define

$$C' = \sup\left\{|\partial^\beta f(x)| \colon x \in K,\ |\beta| \leqslant N\right\}.$$

Then, for any $x \in K$, and $|\alpha| \leqslant N$, we have

$$\left|\partial^\alpha(f\phi)(x)\right| \leqslant \sum_{\substack{\beta+\gamma=\alpha \\ \beta,\gamma \geqslant 0}} \frac{\alpha!}{\beta!\gamma!} C'|\partial^\gamma \phi(x)| \leqslant \left(\sum_{\substack{\beta+\gamma=\alpha \\ \beta,\gamma \geqslant 0}} \frac{\alpha!}{\beta!\gamma!} C'\right)\|\phi\|_N$$

As $\alpha$ ranges over all multi-indices of absolute value at most $N$, we can take supremum of the left hand side to obtain

$$\|f\phi\|_N \leqslant C''\|\phi\|_N,$$

where $C''$ is a constant independent of $\phi$. Therefore,

$$(f\Lambda)(\phi) = \Lambda(f\phi) \leqslant C\|f\phi\|_N \leqslant CC''\|\phi\|_N,$$

and hence, $f\Lambda$ is a distribution.

**Theorem 10.11 (Countable Partition of Unity).** If $\Gamma$ is a c ollection of open sets in $\mathbb{R}^n$ whose union is $\Omega$, then there exists a sequence $\{\psi_i\}$ of elements in $\mathscr{D}(\Omega)$, with $\psi_i \geqslant 0$, such that

(a) each $\psi_i$ is supported in some member of $\Gamma$,

(b) $\sum_i \psi_i(x) = 1$ for every $x \in \Omega$,

(c) to every compact $K \Subset \Omega$, there is an integer $m$ and an open set $W \supseteq K$ such that

$$\psi_1(x) + \cdots + \psi_m(x) = 1 \qquad \forall\ x \in W.$$

Such a collection $\{\psi_i\}$ is called a *locally finite partition of unity* in $\Omega$ *subordinate* to the open cover $\Gamma$.

**Definition 10.12.** Suppose $\Lambda \in \mathscr{D}'(\Omega)$. If $\omega \subset \Omega$ is an open set and if $\Lambda\phi = 0$ for every $\phi \in \mathscr{D}(\Omega)$, we say that $\Lambda$ *vanishes in* $\omega$. Let $W$ be the union of all open $\omega \subseteq \Omega$ in which $\Lambda$ vanishes. The set $\Omega \setminus W$ is the *support* of $\Lambda$.

**Theorem 10.13.** If $W$ is as above, then $\Lambda$ vanishes in $W$.

*Proof.* Let $\Gamma$ be the collection of all $\omega$ as in the above definition. Let $\{\psi_i\}$ be a locally finite partition of unity in $W$, subordinate to $\Gamma$. If $\phi \in \mathscr{D}(W)$, then since $\phi$ has compact support contained in $W$, all but finitely many $\psi_i$ vanish on the support of $\phi$, in particular, we can write

$$\phi = \sum_i \psi_i \phi,$$

where the sum on the right is essentially finite. Thus, we can write

$$\Lambda(\phi) = \sum_i \Lambda(\psi_i \phi),$$

but the support of each $\psi_i$ is contained in some $\omega$ on which $\Lambda$ vanishes. Consequently, the sum on the right is identically 0, as desired. ■

**Theorem 10.14.** Let $\Lambda \in \mathscr{D}'(\Omega)$ and $S_\Lambda$ be the support of $\Lambda$.

(a) If the support of $\phi \in \mathscr{D}(\Omega)$ is disjoint from $S_\Lambda$, then $\Lambda\phi = 0$.

(b) If $S_\Lambda$ is empty, then $\Lambda = 0$.

(c) If $\psi \in C^\infty(\Omega)$ and $\psi = 1$ on some open set $V$ containing $S_\Lambda$, then $\phi\Lambda = \Lambda$.

(d) If $S_\Lambda$ is a compact subset of $\Omega$, then $\Lambda$ has finite order. In fact, there is a constant $C < \infty$ and a nonnegative integer $N$ such that $|\Lambda\phi| \leqslant C\|\phi\|_N$ for every $\phi \in \mathscr{D}(\Omega)$. Further, $\Lambda$ extends in a unique way to a continuous linear functional on $C^\infty(\Omega)$.

*Proof.* (a) This is just a restatement of the preceding result.

(b) Again, this is an immediate consequence of either the preceding result or (a).

(c) Let $\psi \in \mathscr{D}(\Omega)$. Consider the function $\phi - \psi\phi$. This vanishes on $V$, an open set containing $S_\Lambda$, and hence, the support of $\phi - \psi\phi$ is disjoint from $S_\Lambda$. Due to (a), we must have

$$\Lambda\left(\phi - \psi\phi\right) = 0 \implies \Lambda\phi = \Lambda(\psi\phi) = (\psi\Lambda)(\phi),$$

as desired.

(d) In light of Theorem 10.11 (d) with $\Gamma = \{\Omega\}$, there is a $\psi \in \mathscr{D}(\Omega)$ which is identically 1 on an open set $V$ containing $S_\Lambda$. Due to (c), we have $\psi\Lambda = \Lambda$. Let $K$ denote the support of $\psi$. Then, there is a positive integer $N$ and a constant $C > 0$ such that $|\Lambda\phi| \leqslant C\|\phi\|_N$ for all $\phi \in \mathscr{D}_K$. Consequently, for any $\phi \in \mathscr{D}(\Omega)$, we can write

$$|\Lambda\phi| = |\Lambda(\psi\phi)| \leqslant C\|\psi\phi\|_N \leqslant CC'\|\phi\|_N,$$

where the last inequality has been argued earlier while showing that the differentiation of a distribution gives a distribution. It follows that $\Lambda$ is a distribution of finite order and that the constant $CC'$ is independent of the choice of compact set containing the support of $\phi$.

Finally, we must show that there is a unique extension of $\Lambda$ to $C^\infty(\Omega)$. For each $f \in C^\infty(\Omega)$, define

$$\Lambda f = \Lambda(\psi f).$$

This is obviously an extension of $\Lambda$ defined on $\mathscr{D}(\Omega)$. We must show that this is continuous. Indeed, recall that $K$ is the support of $\psi$ and choose an exhaustion $K_0 \subset K_1 \subset \cdots$ of $\Omega$. Choose a positive integer $M$ sufficiently large so that $M \geqslant N$ and $K \subseteq K_M$. Further, using an analogous argument as before, there is a constant $\widetilde{C}$ such that $\rho_M(\psi f) \leqslant \widetilde{C}\rho_M(f)$. Indeed, for $|\alpha| \leqslant M$ and $x \in K_M$, we have

$$\left|\partial^\alpha(f\psi)(x)\right| = \left|\sum_{\substack{\beta+\gamma=\alpha \\ \beta,\gamma\geqslant 0}} \frac{\alpha!}{\beta!\gamma!}\partial^\beta\psi(x)\partial^\gamma f(x)\right| \leqslant \widetilde{C}\rho_M(f).$$

It follows that $\Lambda$ is a continuous linear functional on $C^\infty(\Omega)$. It remains to show that $\mathscr{D}(\Omega)$ is dense in $C^\infty(\Omega)$, whence it would follow that the extension is unique. Indeed, for any $f \in C^\infty(\Omega)$ and positive integer $n$, we can find $\psi_n \in \mathscr{D}(\Omega)$ such that $\psi_n = 1$ on $K_n$. Then, $f - \psi_n f = 0$ on $K_m$ for all $m \leqslant n$. In particular, this means that $p_m(f - \psi_n f) \to 0$ as $n \to \infty$, consequently, $\psi_n f \to f$ in $C^\infty(\Omega)$, as desired. This completes the proof. ■